**1)**

$\mathbf{A}$ and $\mathbf{B}$ are inverse matrices, so $\mathbf{B} = \mathbf{A}^{-1}$ or $\mathbf{A} = \mathbf{B}^{-1}$

**2)**

If $\mathbf{A}$ is singular, then $\det \mathbf{A} = 0$, so $\mathbf{A}$ does not have an inverse matrix as a step in finding the inverse matrix requires you to divide by $\det \mathbf{A}$

**3) a)**

$$\begin{vmatrix} 3 & 7 \\ 2 & 5 \end{vmatrix} = 3 \times 5 - 7 \times 2 = 1$$

$$\begin{pmatrix} 3 & 7 \\ 2 & 5 \end{pmatrix}^{-1} = \frac{1}{1}\begin{pmatrix} 5 & -7 \\ -2 & 3 \end{pmatrix} = \begin{pmatrix} 5 & -7 \\ -2 & 3 \end{pmatrix}$$

**3) b)**

$$\begin{vmatrix} 8 & 13 \\ -3 & -5 \end{vmatrix} = 8 \times (-5) - 13 \times (-3) = -1$$

$$\begin{pmatrix} 8 & 13 \\ -3 & -5 \end{pmatrix}^{-1} = \frac{1}{-1}\begin{pmatrix} -5 & -13 \\ 3 & 8 \end{pmatrix} = \begin{pmatrix} 5 & 13 \\ -3 & -8 \end{pmatrix}$$

**3) c)**

$$\begin{vmatrix} 9 & 1 \\ 28 & 3 \end{vmatrix} = 9 \times 3 - 1 \times 28 = -1$$

$$\begin{pmatrix} 9 & 1 \\ 28 & 3 \end{pmatrix}^{-1} = \frac{1}{-1}\begin{pmatrix} 3 & -1 \\ -28 & 9 \end{pmatrix} = \begin{pmatrix} -3 & 1 \\ 28 & -9 \end{pmatrix}$$

**3) d)**

$$\begin{vmatrix} 6 & 17 \\ 2 & 6 \end{vmatrix} = 6 \times 6 - 17 \times 2 = 2$$

$$\begin{pmatrix} 6 & 17 \\ 2 & 6 \end{pmatrix}^{-1} = \frac{1}{2}\begin{pmatrix} 6 & -17 \\ -2 & 6 \end{pmatrix} = \begin{pmatrix} 3 & -\frac{17}{2} \\ -1 & 3 \end{pmatrix}$$

**3) e)**

$$\begin{vmatrix} -8 & -7 \\ 11 & 10 \end{vmatrix} = -8 \times 10 - (-7) \times 11 = -3$$

$$\begin{pmatrix} -8 & -7 \\ 11 & 10 \end{pmatrix}^{-1} = \frac{1}{-3}\begin{pmatrix} 10 & 7 \\ -11 & -8 \end{pmatrix} = \begin{pmatrix} -\frac{10}{3} & -\frac{7}{3} \\ \frac{11}{3} & \frac{8}{3} \end{pmatrix}$$

**3) f)**

$$\begin{vmatrix} 12 & 20 \\ 5 & 10 \end{vmatrix} = 12 \times 10 - 20 \times 5 = 20$$

$$\begin{pmatrix} 12 & 20 \\ 5 & 10 \end{pmatrix}^{-1} = \frac{1}{20}\begin{pmatrix} 10 & -20 \\ -5 & 12 \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & -1 \\ -\frac{1}{4} & \frac{3}{5} \end{pmatrix}$$

**3) g)**

$$\begin{vmatrix} -2 & 9 \\ 1 & 3 \end{vmatrix} = -2 \times 3 - 9 \times 1 = -15$$

$$\begin{pmatrix} -2 & 9 \\ 1 & 3 \end{pmatrix}^{-1} = \frac{1}{-15}\begin{pmatrix} 3 & -9 \\ -1 & -2 \end{pmatrix} = \begin{pmatrix} -\frac{1}{5} & \frac{3}{5} \\ \frac{1}{15} & \frac{2}{15} \end{pmatrix}$$

**3) h)**

$$\begin{vmatrix} 6 & -5 \\ -25 & 0 \end{vmatrix} = 6 \times 0 - (-5) \times (-25) = -125$$

$$\begin{pmatrix} 6 & -5 \\ -25 & 0 \end{pmatrix}^{-1} = \frac{1}{-125}\begin{pmatrix} 0 & 5 \\ 25 & 6 \end{pmatrix} = \begin{pmatrix} 0 & -\frac{1}{25} \\ -\frac{1}{5} & -\frac{6}{125} \end{pmatrix}$$

**3) i)**

$$\begin{vmatrix} -3 & 8 \\ -9 & 7 \end{vmatrix} = (-3) \times 7 - 8 \times (-9) = 51$$

$$\begin{pmatrix} -3 & 8 \\ -9 & 7 \end{pmatrix}^{-1} = \frac{1}{51}\begin{pmatrix} 7 & -8 \\ 9 & -3 \end{pmatrix} = \begin{pmatrix} \frac{7}{51} & -\frac{8}{51} \\ \frac{3}{17} & -\frac{1}{17} \end{pmatrix}$$

**3) j)**

$$\begin{vmatrix} 20 & -30 \\ -50 & -40 \end{vmatrix} = 20 \times (-40) - (-30) \times (-50) = -2300$$

$$\begin{pmatrix} 20 & -30 \\ -50 & -40 \end{pmatrix}^{-1} = \frac{1}{-2300}\begin{pmatrix} -40 & 30 \\ 50 & 20 \end{pmatrix} = \begin{pmatrix} \frac{2}{115} & -\frac{3}{230} \\ -\frac{1}{46} & -\frac{1}{115} \end{pmatrix}$$

**4) a)**

$$\det \mathbf{A} = 0 \Rightarrow k \times 3 - 4 \times 2 = 0 \Rightarrow 3k - 8 = 0 \Rightarrow k = \frac{8}{3}$$

**4) b)**

$$\begin{pmatrix} k & 4 \\ 2 & 3 \end{pmatrix}^{-1} = \frac{1}{3k-8}\begin{pmatrix} 3 & -4 \\ -2 & k \end{pmatrix} = \begin{pmatrix} \frac{3}{3k-8} & -\frac{4}{3k-8} \\ -\frac{2}{3k-8} & \frac{k}{3k-8} \end{pmatrix}$$

**5) a)**

$\det \mathbf{A} = 0 \Rightarrow k \times k - 2 \times 2 = 0 \Rightarrow k^2 - 4 = 0 \Rightarrow k = \pm 2$

As $k$ is a positive constant, $k = 2$

**5) b)**

$$\begin{pmatrix} k & 2 \\ 2 & k \end{pmatrix}^{-1} = \frac{1}{k^2-4}\begin{pmatrix} k & -2 \\ -2 & k \end{pmatrix} = \begin{pmatrix} \frac{k}{k^2-4} & -\frac{2}{k^2-4} \\ -\frac{2}{k^2-4} & \frac{k}{k^2-4} \end{pmatrix}$$

**6) a)**

$\det \mathbf{A} = 0 \Rightarrow (k+1) \times (k-1) - \left(-2\sqrt{2}\right) \times \left(-2\sqrt{2}\right) = 0 \Rightarrow k^2 - 1 - 8 = 0$

$\Rightarrow k^2 - 9 = 0 \Rightarrow k = \pm 3$

As $k$ is a negative constant, $k = -3$

**6) b)**

$$\begin{pmatrix} k+1 & -2\sqrt{2} \\ -2\sqrt{2} & k-1 \end{pmatrix}^{-1} = \frac{1}{k^2-9}\begin{pmatrix} k-1 & 2\sqrt{2} \\ 2\sqrt{2} & k+1 \end{pmatrix} = \begin{pmatrix} \frac{k-1}{k^2-9} & \frac{2\sqrt{2}}{k^2-9} \\ \frac{2\sqrt{2}}{k^2-9} & \frac{k+1}{k^2-9} \end{pmatrix}$$

**7) a) i)**

Enlargement, centred at the origin, scale factor 3

**Alternative Method:**

Stretch, parallel to the $x$-axis, scale factor 3, and a Stretch, parallel to the $y$-axis, scale factor 3

**7) a) ii)**

$$\begin{vmatrix} 3 & 0 \\ 0 & 3 \end{vmatrix} = 3 \times 3 - 0 \times 0 = 9$$

$$\begin{pmatrix} 3 & 0 \\ 0 & 3 \end{pmatrix}^{-1} = \frac{1}{9}\begin{pmatrix} 3 & 0 \\ 0 & 3 \end{pmatrix} = \begin{pmatrix} \frac{1}{3} & 0 \\ 0 & \frac{1}{3} \end{pmatrix}$$

**7) a) iii)**

Enlargement, centred at the origin, scale factor $\frac{1}{3}$

**Alternative Method:**

Stretch, parallel to the $x$-axis, scale factor $\frac{1}{3}$, and a Stretch, parallel to the $y$-axis, scale factor $\frac{1}{3}$

**7) b) i)**

Reflection in the line $y = x$

**7) b) ii)**

$$\begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix} = 0 \times 0 - 1 \times 1 = -1$$

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}^{-1} = \frac{1}{-1}\begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

**7) b) iii)**

Reflection in the line $y = x$

**7) c) i)**

Stretch, parallel to the $x$-axis, scale factor $8$

**7) c) ii)**

$$\begin{vmatrix} 8 & 0 \\ 0 & 1 \end{vmatrix} = 8 \times 1 - 0 \times 0 = 8$$

$$\begin{pmatrix} 8 & 0 \\ 0 & 1 \end{pmatrix}^{-1} = \frac{1}{8}\begin{pmatrix} 1 & 0 \\ 0 & 8 \end{pmatrix} = \begin{pmatrix} \frac{1}{8} & 0 \\ 0 & 1 \end{pmatrix}$$

**7) c) iii)**

Stretch, parallel to the $x$-axis, scale factor $\frac{1}{8}$

**7) d) i)**

Rotation, anticlockwise about the origin, by $90°$

**Alternative Method:**

Rotation, clockwise about the origin, by $270°$

**7) d) ii)**

$$\begin{vmatrix} 0 & -1 \\ 1 & 0 \end{vmatrix} = 0 \times 0 - (-1) \times 1 = 1$$

$$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}^{-1} = \frac{1}{1}\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

**7) d) iii)**

Rotation, anticlockwise about the origin, by $270°$

**Alternative Method:**

Rotation, clockwise about the origin, by $90°$

**7) e) i)**

Reflection in the $x$-axis

**7) e) ii)**

$$\begin{vmatrix} 1 & 0 \\ 0 & -1 \end{vmatrix} = 1 \times (-1) - 0 \times 0 = -1$$

$$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}^{-1} = \frac{1}{-1}\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

**7) e) iii)**

Reflection in the $x$-axis

**7) f) i)**

Rotation, about the origin, by $180°$

**7) f) ii)**

$$\begin{vmatrix} -1 & 0 \\ 0 & -1 \end{vmatrix} = (-1) \times (-1) - 0 \times 0 = 1$$

$$\begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}^{-1} = \frac{1}{1}\begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$$

**7) f) iii)**

Rotation, about the origin, by $180°$

**7) g) i)**

Rotation, anticlockwise about the origin, by $270°$

**Alternative Method:**

Rotation, clockwise about the origin, by $90°$

**7) g) ii)**

$$\begin{vmatrix} 0 & 1 \\ -1 & 0 \end{vmatrix} = 0 \times 0 - 1 \times (-1) = 1$$

$$\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}^{-1} = \frac{1}{1}\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

**7) g) iii)**

Rotation, anticlockwise about the origin, by $90°$

**Alternative Method:**

Rotation, clockwise about the origin, by $270°$

**7) h) i)**

Reflection in the $y$-axis

**7) h) ii)**

$$\begin{vmatrix} -1 & 0 \\ 0 & 1 \end{vmatrix} = (-1) \times 1 - 0 \times 0 = -1$$

$$\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}^{-1} = \frac{1}{-1}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$$

**7) h) iii)**

Reflection in the $y$-axis

**7) i) i)**

Rotation, anticlockwise about the origin, by $45°$

**Alternative Method:**

Rotation, clockwise about the origin, by $315°$

**7) i) ii)**

$$\begin{vmatrix} \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{vmatrix} = \frac{\sqrt{2}}{2} \times \frac{\sqrt{2}}{2} - \left(-\frac{\sqrt{2}}{2}\right) \times \frac{\sqrt{2}}{2} = 1$$

$$\begin{pmatrix} \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{pmatrix}^{-1} = \frac{1}{1}\begin{pmatrix} \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{pmatrix} = \begin{pmatrix} \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{pmatrix}$$

**7) i) iii)**

Rotation, anticlockwise about the origin, by $315°$

**Alternative Method:**

Rotation, clockwise about the origin, by $45°$

**7) j) i)**

Rotation, anticlockwise about the origin, by $225°$

**Alternative Method:**

Rotation, clockwise about the origin, by $135°$

**7) j) ii)**

$$\begin{vmatrix} -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ -\frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \end{vmatrix} = \left(-\frac{\sqrt{2}}{2}\right) \times \left(-\frac{\sqrt{2}}{2}\right) - \frac{\sqrt{2}}{2} \times \left(-\frac{\sqrt{2}}{2}\right) = 1$$

$$\begin{pmatrix} -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ -\frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \end{pmatrix}^{-1} = \frac{1}{1}\begin{pmatrix} -\frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \end{pmatrix} = \begin{pmatrix} -\frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \end{pmatrix}$$

**7) j) iii)**

Rotation, anticlockwise about the origin, by $135°$

**Alternative Method:**

Rotation, clockwise about the origin, by $225°$

**7) k) i)**

Rotation, anticlockwise about the origin, by $30°$

**Alternative Method:**

Rotation, clockwise about the origin, by $330°$

**7) k) ii)**

$$\begin{vmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{\sqrt{3}}{2} \end{vmatrix} = \frac{\sqrt{3}}{2} \times \frac{\sqrt{3}}{2} - \left(-\frac{1}{2}\right) \times \frac{1}{2} = 1$$

$$\begin{pmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{\sqrt{3}}{2} \end{pmatrix}^{-1} = \frac{1}{1}\begin{pmatrix} \frac{\sqrt{3}}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{\sqrt{3}}{2} \end{pmatrix} = \begin{pmatrix} \frac{\sqrt{3}}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{\sqrt{3}}{2} \end{pmatrix}$$

**7) k) iii)**
Rotation, anticlockwise about the origin, by $330°$
**Alternative Method:**
Rotation, clockwise about the origin, by $30°$

**7) l) i)**
Rotation, anticlockwise about the origin, by $300°$
**Alternative Method:**
Rotation, clockwise about the origin, by $60°$

**7) l) ii)**
$$\begin{vmatrix} \frac{1}{2} & \frac{\sqrt{3}}{2} \\ -\frac{\sqrt{3}}{2} & \frac{1}{2} \end{vmatrix} = \frac{1}{2} \times \frac{1}{2} - \frac{\sqrt{3}}{2} \times \left(-\frac{\sqrt{3}}{2}\right) = 1$$
$$\begin{pmatrix} \frac{1}{2} & \frac{\sqrt{3}}{2} \\ -\frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix}^{-1} = \frac{1}{1}\begin{pmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix}$$

**7) l) iii)**
Rotation, anticlockwise about the origin, by $60°$
**Alternative Method:**
Rotation, clockwise about the origin, by $300°$

**8) a) i)**
Shear, with the $x$-axis fixed, with $(0, 1)$ mapped to $(4, 1)$

**8) a) ii)**
$$\begin{vmatrix} 1 & 4 \\ 0 & 1 \end{vmatrix} = 1 \times 1 - 4 \times 0 = 1$$
$$\begin{pmatrix} 1 & 4 \\ 0 & 1 \end{pmatrix}^{-1} = \frac{1}{1}\begin{pmatrix} 1 & -4 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & -4 \\ 0 & 1 \end{pmatrix}$$

**8) a) iii)**
Shear, with the $x$-axis fixed, with $(0, 1)$ mapped to $(-4, 1)$

**8) b) i)**
Shear, with the $y$-axis fixed, with $(1, 0)$ mapped to $(1, 6)$

**8) b) ii)**

$$\begin{vmatrix} 1 & 0 \\ 6 & 1 \end{vmatrix} = 1 \times 1 - 0 \times 6 = 1$$

$$\begin{pmatrix} 1 & 0 \\ 6 & 1 \end{pmatrix}^{-1} = \frac{1}{1}\begin{pmatrix} 1 & 0 \\ -6 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ -6 & 1 \end{pmatrix}$$

**8) b) iii)**

Shear, with the $y$-axis fixed, with $(1, 0)$ mapped to $(1, -6)$

**8) c) i)**

Shear, with the $x$-axis fixed, with $(0, 1)$ mapped to $(-2, 1)$

**8) c) ii)**

$$\begin{vmatrix} 1 & -2 \\ 0 & 1 \end{vmatrix} = 1 \times 1 - (-2) \times 0 = 1$$

$$\begin{pmatrix} 1 & -2 \\ 0 & 1 \end{pmatrix}^{-1} = \frac{1}{1}\begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$$

**8) c) iii)**

Shear, with the $x$-axis fixed, with $(0, 1)$ mapped to $(2, 1)$

**8) d) i)**

Shear, with the $y$-axis fixed, with $(1, 0)$ mapped to $\left(1, -\frac{2}{5}\right)$

**8) d) ii)**

$$\begin{vmatrix} 1 & 0 \\ -\frac{2}{5} & 1 \end{vmatrix} = 1 \times 1 - 0 \times \left(-\frac{2}{5}\right) = 1$$

$$\begin{pmatrix} 1 & 0 \\ -\frac{2}{5} & 1 \end{pmatrix}^{-1} = \frac{1}{1}\begin{pmatrix} 1 & 0 \\ \frac{2}{5} & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ \frac{2}{5} & 1 \end{pmatrix}$$

**8) d) iii)**

Shear, with the $y$-axis fixed, with $(1, 0)$ mapped to $\left(1, \frac{2}{5}\right)$

**9) a)**

$$\mathbf{A} = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$$

**9) b)**

$$\mathbf{B} = \begin{pmatrix} \cos(-\theta) & -\sin(-\theta) \\ \sin(-\theta) & \cos(-\theta) \end{pmatrix} = \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix}$$

**9) c)**

$$\mathbf{AB} = \mathbf{I} \Rightarrow \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}\begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} \cos^2\theta + \sin^2\theta & \sin\theta\cos\theta - \sin\theta\cos\theta \\ \sin\theta\cos\theta - \sin\theta\cos\theta & \sin^2\theta + \cos^2\theta \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} \cos^2\theta + \sin^2\theta & 0 \\ 0 & \sin^2\theta + \cos^2\theta \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$\therefore \sin^2\theta + \cos^2\theta \equiv 1$$

**10) a)**

$$\det\mathbf{A} = 2 \times 5 - 3 \times 3 = 1$$

$$\mathbf{A}^{-1} = \frac{1}{1}\begin{pmatrix} 5 & -3 \\ -3 & 2 \end{pmatrix} = \begin{pmatrix} 5 & -3 \\ -3 & 2 \end{pmatrix}$$

**10) b)**

Let the coordinates of $P$ be $(a, b)$

$$\mathbf{A}\begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 1 \\ 3 \end{pmatrix} \Rightarrow \mathbf{A}^{-1}\mathbf{A}\begin{pmatrix} a \\ b \end{pmatrix} = \mathbf{A}^{-1}\begin{pmatrix} 1 \\ 3 \end{pmatrix} \Rightarrow \begin{pmatrix} a \\ b \end{pmatrix} = \mathbf{A}^{-1}\begin{pmatrix} 1 \\ 3 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 5 & -3 \\ -3 & 2 \end{pmatrix}\begin{pmatrix} 1 \\ 3 \end{pmatrix} = \begin{pmatrix} -4 \\ 3 \end{pmatrix}$$

So $P$ has coordinates $(-4, 3)$

**11) a)**

$$\det\mathbf{A} = \frac{1}{\sqrt{2}} \times \frac{1}{\sqrt{2}} - \left(-\frac{1}{\sqrt{2}}\right) \times \frac{1}{\sqrt{2}} = 1$$

$$\mathbf{A}^{-1} = \frac{1}{1}\begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix} = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix}$$

**11) b)**

Let the coordinates of $P$ be $(a, b)$

$$\mathbf{A}\begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} -3 \\ 1 \end{pmatrix} \Rightarrow \mathbf{A}^{-1}\mathbf{A}\begin{pmatrix} a \\ b \end{pmatrix} = \mathbf{A}^{-1}\begin{pmatrix} -3 \\ 1 \end{pmatrix} \Rightarrow \begin{pmatrix} a \\ b \end{pmatrix} = \mathbf{A}^{-1}\begin{pmatrix} -3 \\ 1 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix}\begin{pmatrix} -3 \\ 1 \end{pmatrix} = \begin{pmatrix} -\sqrt{2} \\ 2\sqrt{2} \end{pmatrix}$$

So $P$ has coordinates $\left(-\sqrt{2}, 2\sqrt{2}\right)$

**12)**

$\mathbf{AB} = \mathbf{C} \Rightarrow \mathbf{A}^{-1}\mathbf{AB} = \mathbf{A}^{-1}\mathbf{C} \Rightarrow \mathbf{IB} = \mathbf{A}^{-1}\mathbf{C} \Rightarrow \mathbf{B} = \mathbf{A}^{-1}\mathbf{C}$

$\det \mathbf{A} = 3 \times 1 - 2 \times 1 = 1$

$\mathbf{A}^{-1} = \frac{1}{1}\begin{pmatrix} 1 & -2 \\ -1 & 3 \end{pmatrix} = \begin{pmatrix} 1 & -2 \\ -1 & 3 \end{pmatrix}$

$\therefore \mathbf{B} = \begin{pmatrix} 1 & -2 \\ -1 & 3 \end{pmatrix}\begin{pmatrix} 2 & 3 \\ 1 & 2 \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 1 & 3 \end{pmatrix}$

**13)**

$\mathbf{AB} = \mathbf{C} \Rightarrow \mathbf{ABB}^{-1} = \mathbf{CB}^{-1} \Rightarrow \mathbf{AI} = \mathbf{CB}^{-1} \Rightarrow \mathbf{A} = \mathbf{CB}^{-1}$

$\det \mathbf{B} = 4 \times 5 - 3 \times 6 = 2$

$\mathbf{B}^{-1} = \frac{1}{2}\begin{pmatrix} 5 & -3 \\ -6 & 4 \end{pmatrix} = \begin{pmatrix} \frac{5}{2} & -\frac{3}{2} \\ -3 & 2 \end{pmatrix}$

$\therefore \mathbf{A} = \begin{pmatrix} 6 & 2 \\ 4 & 1 \end{pmatrix}\begin{pmatrix} \frac{5}{2} & -\frac{3}{2} \\ -3 & 2 \end{pmatrix} = \begin{pmatrix} 9 & -5 \\ 7 & -4 \end{pmatrix}$

**14)**

$\mathbf{AB} = \begin{pmatrix} 2 & -1 & 3 \\ -2 & -4 & 5 \end{pmatrix} \Rightarrow \begin{pmatrix} 3 & 1 \\ 4 & 2 \end{pmatrix}\mathbf{B} = \begin{pmatrix} 2 & -1 & 3 \\ -2 & -4 & 5 \end{pmatrix}$

$\Rightarrow \begin{pmatrix} 3 & 1 \\ 4 & 2 \end{pmatrix}^{-1}\begin{pmatrix} 3 & 1 \\ 4 & 2 \end{pmatrix}\mathbf{B} = \begin{pmatrix} 3 & 1 \\ 4 & 2 \end{pmatrix}^{-1}\begin{pmatrix} 2 & -1 & 3 \\ -2 & -4 & 5 \end{pmatrix}$

$\Rightarrow \mathbf{IB} = \begin{pmatrix} 3 & 1 \\ 4 & 2 \end{pmatrix}^{-1}\begin{pmatrix} 2 & -1 & 3 \\ -2 & -4 & 5 \end{pmatrix} \Rightarrow \mathbf{B} = \begin{pmatrix} 3 & 1 \\ 4 & 2 \end{pmatrix}^{-1}\begin{pmatrix} 2 & -1 & 3 \\ -2 & -4 & 5 \end{pmatrix}$

$\begin{vmatrix} 3 & 1 \\ 4 & 2 \end{vmatrix} = 3 \times 2 - 1 \times 4 = 2$

$\begin{pmatrix} 3 & 1 \\ 4 & 2 \end{pmatrix}^{-1} = \frac{1}{2}\begin{pmatrix} 2 & -1 \\ -4 & 3 \end{pmatrix} = \begin{pmatrix} 1 & -\frac{1}{2} \\ -2 & \frac{3}{2} \end{pmatrix}$

$\therefore \mathbf{B} = \begin{pmatrix} 1 & -\frac{1}{2} \\ -2 & \frac{3}{2} \end{pmatrix}\begin{pmatrix} 2 & -1 & 3 \\ -2 & -4 & 5 \end{pmatrix} = \begin{pmatrix} 3 & 1 & \frac{1}{2} \\ -7 & -4 & \frac{3}{2} \end{pmatrix}$

**15)**

$$\mathbf{AB}=\begin{pmatrix}34 & -1\\ -24 & 11\\ 16 & 1\end{pmatrix}\Rightarrow \mathbf{A}\begin{pmatrix}9 & -1\\ -2 & 3\end{pmatrix}=\begin{pmatrix}34 & -1\\ -24 & 11\\ 16 & 1\end{pmatrix}$$

$$\Rightarrow \mathbf{A}\begin{pmatrix}9 & -1\\ -2 & 3\end{pmatrix}\begin{pmatrix}9 & -1\\ -2 & 3\end{pmatrix}^{-1}=\begin{pmatrix}34 & -1\\ -24 & 11\\ 16 & 1\end{pmatrix}\begin{pmatrix}9 & -1\\ -2 & 3\end{pmatrix}^{-1}$$

$$\Rightarrow \mathbf{AI}=\begin{pmatrix}34 & -1\\ -24 & 11\\ 16 & 1\end{pmatrix}\begin{pmatrix}9 & -1\\ -2 & 3\end{pmatrix}^{-1}$$

$$\Rightarrow \mathbf{A}=\begin{pmatrix}34 & -1\\ -24 & 11\\ 16 & 1\end{pmatrix}\begin{pmatrix}9 & -1\\ -2 & 3\end{pmatrix}^{-1}$$

$$\begin{vmatrix}9 & -1\\ -2 & 3\end{vmatrix}=9\times 3-(-1)\times(-2)=25$$

$$\begin{pmatrix}9 & -1\\ -2 & 3\end{pmatrix}^{-1}=\frac{1}{25}\begin{pmatrix}3 & 1\\ 2 & 9\end{pmatrix}=\begin{pmatrix}\frac{3}{25} & \frac{1}{25}\\ \frac{2}{25} & \frac{9}{25}\end{pmatrix}$$

$$\therefore \mathbf{A}=\begin{pmatrix}34 & -1\\ -24 & 11\\ 16 & 1\end{pmatrix}\begin{pmatrix}\frac{3}{25} & \frac{1}{25}\\ \frac{2}{25} & \frac{9}{25}\end{pmatrix}=\begin{pmatrix}4 & 1\\ -2 & 3\\ 2 & 1\end{pmatrix}$$

**16)**

$$\mathbf{AB}=\begin{pmatrix}9 & 15\\ 33 & 39\\ 11 & 21\\ 26 & 54\end{pmatrix}\Rightarrow \mathbf{A}\begin{pmatrix}5 & 3\\ 2 & 6\end{pmatrix}=\begin{pmatrix}9 & 15\\ 33 & 39\\ 11 & 21\\ 26 & 54\end{pmatrix}$$

$$\Rightarrow \mathbf{A}\begin{pmatrix}5 & 3\\ 2 & 6\end{pmatrix}\begin{pmatrix}5 & 3\\ 2 & 6\end{pmatrix}^{-1}=\begin{pmatrix}9 & 15\\ 33 & 39\\ 11 & 21\\ 26 & 54\end{pmatrix}\begin{pmatrix}5 & 3\\ 2 & 6\end{pmatrix}^{-1}$$

$$\Rightarrow \mathbf{AI}=\begin{pmatrix}9 & 15\\ 33 & 39\\ 11 & 21\\ 26 & 54\end{pmatrix}\begin{pmatrix}5 & 3\\ 2 & 6\end{pmatrix}^{-1}\Rightarrow \mathbf{A}=\begin{pmatrix}9 & 15\\ 33 & 39\\ 11 & 21\\ 26 & 54\end{pmatrix}\begin{pmatrix}5 & 3\\ 2 & 6\end{pmatrix}^{-1}$$

$$\begin{vmatrix}5 & 3\\ 2 & 6\end{vmatrix}=5\times 6-3\times 2=24$$

$$\begin{pmatrix}5 & 3\\ 2 & 6\end{pmatrix}^{-1}=\frac{1}{24}\begin{pmatrix}6 & -3\\ -2 & 5\end{pmatrix}=\begin{pmatrix}\frac{1}{4} & -\frac{1}{8}\\ -\frac{1}{12} & \frac{5}{24}\end{pmatrix}$$

$$\therefore \mathbf{A}=\begin{pmatrix}9 & 15\\ 33 & 39\\ 11 & 21\\ 26 & 54\end{pmatrix}\begin{pmatrix}\frac{1}{4} & -\frac{1}{8}\\ -\frac{1}{12} & \frac{5}{24}\end{pmatrix}=\begin{pmatrix}1 & 2\\ 5 & 4\\ 1 & 3\\ 2 & 8\end{pmatrix}$$

**17) a)**

$\det \mathbf{A} = 2 \times 8 - 4 \times (-9) = 52$

$\mathbf{A}^{-1} = \frac{1}{52}\begin{pmatrix} 8 & -4 \\ 9 & 2 \end{pmatrix} = \begin{pmatrix} \frac{2}{13} & -\frac{1}{13} \\ \frac{9}{52} & \frac{1}{26} \end{pmatrix}$

**17) b)**

$\mathbf{AB} = \begin{pmatrix} 2 & -2 \\ -35 & 61 \end{pmatrix} \Rightarrow \mathbf{A}^{-1}\mathbf{AB} = \mathbf{A}^{-1}\begin{pmatrix} 2 & -2 \\ -35 & 61 \end{pmatrix} \Rightarrow \mathbf{IB} = \mathbf{A}^{-1}\begin{pmatrix} 2 & -2 \\ -35 & 61 \end{pmatrix}$

$\Rightarrow \mathbf{B} = \mathbf{A}^{-1}\begin{pmatrix} 2 & -2 \\ -35 & 61 \end{pmatrix}$

$\Rightarrow \mathbf{B} = \begin{pmatrix} \frac{2}{13} & -\frac{1}{13} \\ \frac{9}{52} & \frac{1}{26} \end{pmatrix}\begin{pmatrix} 2 & -2 \\ -35 & 61 \end{pmatrix} = \begin{pmatrix} 3 & -5 \\ -1 & 2 \end{pmatrix}$

$\mathbf{C} = 3\mathbf{A} - 4\mathbf{B} = 3\begin{pmatrix} 2 & 4 \\ -9 & 8 \end{pmatrix} - 4\begin{pmatrix} 3 & -5 \\ -1 & 2 \end{pmatrix} = \begin{pmatrix} 6 & 12 \\ -27 & 24 \end{pmatrix} - \begin{pmatrix} 12 & -20 \\ -4 & 8 \end{pmatrix} = \begin{pmatrix} -6 & 32 \\ -23 & 16 \end{pmatrix}$

**18)**

$\mathbf{AA} = \begin{pmatrix} 2 & 3 \\ -1 & -2 \end{pmatrix}\begin{pmatrix} 2 & 3 \\ -1 & -2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

Hence the matrix $A$ is self-inverse

**19)**

Let triangle $T$ have vertices at $(a, b)$, $(c, d)$ and $(e, f)$, so that when $T$ is mapped to $T'$, the vertices $(a, b)$, $(c, d)$ and $(e, f)$ are mapped to $(2, 3)$, $(4, -5)$ and $(-1, -1)$ respectively.

$\begin{pmatrix} 2 & 1 \\ 13 & 7 \end{pmatrix}\begin{pmatrix} a & c & e \\ b & d & f \end{pmatrix} = \begin{pmatrix} 2 & 4 & -1 \\ 3 & -5 & -1 \end{pmatrix}$

$\Rightarrow \begin{pmatrix} 2 & 1 \\ 13 & 7 \end{pmatrix}^{-1}\begin{pmatrix} 2 & 1 \\ 13 & 7 \end{pmatrix}\begin{pmatrix} a & c & e \\ b & d & f \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 13 & 7 \end{pmatrix}^{-1}\begin{pmatrix} 2 & 4 & -1 \\ 3 & -5 & -1 \end{pmatrix}$

$\Rightarrow \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} a & c & e \\ b & d & f \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 13 & 7 \end{pmatrix}^{-1}\begin{pmatrix} 2 & 4 & -1 \\ 3 & -5 & -1 \end{pmatrix}$

$\Rightarrow \begin{pmatrix} a & c & e \\ b & d & f \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 13 & 7 \end{pmatrix}^{-1}\begin{pmatrix} 2 & 4 & -1 \\ 3 & -5 & -1 \end{pmatrix}$

$\begin{vmatrix} 2 & 1 \\ 13 & 7 \end{vmatrix} = 2 \times 7 - 1 \times 13 = 1$

$\begin{pmatrix} 2 & 1 \\ 13 & 7 \end{pmatrix}^{-1} = \frac{1}{1}\begin{pmatrix} 7 & -1 \\ -13 & 2 \end{pmatrix} = \begin{pmatrix} 7 & -1 \\ -13 & 2 \end{pmatrix}$

$\therefore \begin{pmatrix} a & c & e \\ b & d & f \end{pmatrix} = \begin{pmatrix} 7 & -1 \\ -13 & 2 \end{pmatrix}\begin{pmatrix} 2 & 4 & -1 \\ 3 & -5 & -1 \end{pmatrix} = \begin{pmatrix} 11 & 33 & -6 \\ -20 & -62 & 11 \end{pmatrix}$

So the vertices of $T$ are $(11, -20)$, $(33, -62)$ and $(-6, 11)$

**20)**

$\mathbf{AB} = \mathbf{C} \Rightarrow \mathbf{A}^{-1}\mathbf{AB} = \mathbf{A}^{-1}\mathbf{C} \Rightarrow \mathbf{IB} = \mathbf{A}^{-1}\mathbf{C} \Rightarrow \mathbf{B} = \mathbf{A}^{-1}\mathbf{C}$

$\det \mathbf{A} = p \times p - 4q \times 2q = p^2 - 8q^2$

$\mathbf{A}^{-1} = \frac{1}{p^2-8q^2}\begin{pmatrix} p & -2q \\ -4q & p \end{pmatrix}$

$\therefore \mathbf{B} = \frac{1}{p^2-8q^2}\begin{pmatrix} p & -2q \\ -4q & p \end{pmatrix}\begin{pmatrix} 3p & q \\ 5q & 2p \end{pmatrix} = \frac{1}{p^2-8q^2}\begin{pmatrix} 3p^2-10q^2 & pq-4pq \\ -12pq+5pq & -4q^2+2p^2 \end{pmatrix}$

$= \frac{1}{p^2-8q^2}\begin{pmatrix} 3p^2-10q^2 & -3pq \\ -7pq & 2p^2-4q^2 \end{pmatrix} = \begin{pmatrix} \frac{3p^2-10q^2}{p^2-8q^2} & -\frac{3pq}{p^2-8q^2} \\ -\frac{7pq}{p^2-8q^2} & \frac{2p^2-4q^2}{p^2-8q^2} \end{pmatrix}$

**21)**

$\mathbf{AB} = \mathbf{C} \Rightarrow \det(\mathbf{AB}) = \det \mathbf{C} \Rightarrow \det \mathbf{A} \times \det \mathbf{B} = \det \mathbf{C}$

$\Rightarrow \det \mathbf{B} = \frac{\det \mathbf{C}}{\det \mathbf{A}} = \frac{5p \times p - (-2q) \times (-4q)}{3p \times (-p) - 4q \times 2q} = \frac{5p^2-8q^2}{-3p^2-8q^2}$

**Alternative Method:**

$\mathbf{AB} = \mathbf{C} \Rightarrow \mathbf{A}^{-1}\mathbf{AB} = \mathbf{A}^{-1}\mathbf{C} \Rightarrow \mathbf{IB} = \mathbf{A}^{-1}\mathbf{C} \Rightarrow \mathbf{B} = \mathbf{A}^{-1}\mathbf{C}$

$\det \mathbf{A} = 3p \times (-p) - 4q \times 2q = -3p^2 - 8q^2$

$\mathbf{A}^{-1} = \frac{1}{-3p^2-8q^2}\begin{pmatrix} -p & -4q \\ -2q & 3p \end{pmatrix}$

$\therefore \mathbf{B} = \frac{1}{-3p^2-8q^2}\begin{pmatrix} -p & -4q \\ -2q & 3p \end{pmatrix}\begin{pmatrix} 5p & -2q \\ -4q & p \end{pmatrix} = \frac{1}{-3p^2-8q^2}\begin{pmatrix} -5p^2+16q^2 & 2pq-4pq \\ -10pq-12pq & 4q^2+3p^2 \end{pmatrix}$

$= \frac{1}{-3p^2-8q^2}\begin{pmatrix} -5p^2+16q^2 & -2pq \\ -22pq & 3p^2+4q^2 \end{pmatrix}$

$\det \mathbf{B} = \left(\frac{1}{-3p^2-8q^2}\right)^2 \times \begin{vmatrix} -5p^2+16q^2 & -2pq \\ -22pq & 3p^2+4q^2 \end{vmatrix}$

$= \left(\frac{1}{-3p^2-8q^2}\right)^2 \times \left((-5p^2+16q^2) \times (3p^2+4q^2) - (-2pq) \times (-22pq)\right)$

$= \left(\frac{1}{-3p^2-8q^2}\right)^2 \times (-15p^4 - 20p^2q^2 + 48p^2q^2 + 64q^4 - 44p^2q^2)$

$= \frac{-15p^4-16p^2q^2+64q^4}{(-3p^2-8q^2)^2} = \frac{(-3p^2-8q^2)(5p^2-8q^2)}{(-3p^2-8q^2)^2} = \frac{5p^2-8q^2}{-3p^2-8q^2}$

**22)**

$\begin{pmatrix}3k & -5\\ -k & 7\end{pmatrix}\begin{pmatrix}x\\ y\end{pmatrix}=\begin{pmatrix}6k\\ -3\end{pmatrix}\Rightarrow\begin{pmatrix}3k & -5\\ -k & 7\end{pmatrix}^{-1}\begin{pmatrix}3k & -5\\ -k & 7\end{pmatrix}\begin{pmatrix}x\\ y\end{pmatrix}=\begin{pmatrix}3k & -5\\ -k & 7\end{pmatrix}^{-1}\begin{pmatrix}6k\\ -3\end{pmatrix}$

$\Rightarrow\begin{pmatrix}1 & 0\\ 0 & 1\end{pmatrix}\begin{pmatrix}x\\ y\end{pmatrix}=\begin{pmatrix}3k & -5\\ -k & 7\end{pmatrix}^{-1}\begin{pmatrix}6k\\ -3\end{pmatrix}\Rightarrow\begin{pmatrix}x\\ y\end{pmatrix}=\begin{pmatrix}3k & -5\\ -k & 7\end{pmatrix}^{-1}\begin{pmatrix}6k\\ -3\end{pmatrix}$

$\begin{vmatrix}3k & -5\\ -k & 7\end{vmatrix}=3k\times7-(-5)\times(-k)=21k-5k=16k$

$\begin{pmatrix}3k & -5\\ -k & 7\end{pmatrix}^{-1}=\frac{1}{16k}\begin{pmatrix}7 & 5\\ k & 3k\end{pmatrix}$

$\therefore\begin{pmatrix}x\\ y\end{pmatrix}=\frac{1}{16k}\begin{pmatrix}7 & 5\\ k & 3k\end{pmatrix}\begin{pmatrix}6k\\ -3\end{pmatrix}=\frac{1}{16k}\begin{pmatrix}42k-15\\ 6k^2-9k\end{pmatrix}=\begin{pmatrix}\frac{42k-15}{16k}\\ \frac{6k-9}{16}\end{pmatrix}$

$\therefore x=\frac{42k-15}{16k}$ and $y=\frac{6k-9}{16}$

**23)**

$2\mathbf{A}^{-1}+\mathbf{A}=7\mathbf{I}\Rightarrow2\begin{pmatrix}3 & k\\ -2 & 4\end{pmatrix}^{-1}+\begin{pmatrix}3 & k\\ -2 & 4\end{pmatrix}=\begin{pmatrix}7 & 0\\ 0 & 7\end{pmatrix}$

$\Rightarrow2\times\frac{1}{12+2k}\begin{pmatrix}4 & -k\\ 2 & 3\end{pmatrix}+\begin{pmatrix}3 & k\\ -2 & 4\end{pmatrix}=\begin{pmatrix}7 & 0\\ 0 & 7\end{pmatrix}$

$\Rightarrow\frac{1}{6+k}\begin{pmatrix}4 & -k\\ 2 & 3\end{pmatrix}+\begin{pmatrix}3 & k\\ -2 & 4\end{pmatrix}=\begin{pmatrix}7 & 0\\ 0 & 7\end{pmatrix}$

$\Rightarrow\begin{pmatrix}4 & -k\\ 2 & 3\end{pmatrix}+(6+k)\begin{pmatrix}3 & k\\ -2 & 4\end{pmatrix}=(6+k)\begin{pmatrix}7 & 0\\ 0 & 7\end{pmatrix}$

$\Rightarrow\begin{pmatrix}4 & -k\\ 2 & 3\end{pmatrix}+\begin{pmatrix}18+3k & 6k+k^2\\ -12-2k & 24+4k\end{pmatrix}=\begin{pmatrix}42+7k & 0\\ 0 & 42+7k\end{pmatrix}$

$\Rightarrow\begin{pmatrix}22+3k & 5k+k^2\\ -10-2k & 27+4k\end{pmatrix}=\begin{pmatrix}42+7k & 0\\ 0 & 42+7k\end{pmatrix}$

$22+3k=42+7k\Rightarrow-20=4k\Rightarrow k=-5$

$5k+k^2=0\Rightarrow k(5+k)=0\Rightarrow k=0$ or $k=-5$

$-10-2k=0\Rightarrow k=-5$

$27+4k=42+7k\Rightarrow-15=3k\Rightarrow k=-5$

$\therefore k=-5$

**24) a)**

$\det\mathbf{A}=5\times3-7\times2=1\neq0$

$\det\mathbf{B}=4\times6-3\times7=3\neq0$

$\therefore$ $\mathbf{A}$ and $\mathbf{B}$ are non-singular

**24) b)**

$\mathbf{AB} = \begin{pmatrix} 5 & 7 \\ 2 & 3 \end{pmatrix}\begin{pmatrix} 4 & 3 \\ 7 & 6 \end{pmatrix} = \begin{pmatrix} 69 & 57 \\ 29 & 24 \end{pmatrix}$

$\det(\mathbf{AB}) = 69 \times 24 - 57 \times 29 = 3$

$(\mathbf{AB})^{-1} = \frac{1}{3}\begin{pmatrix} 24 & -57 \\ -29 & 69 \end{pmatrix} = \begin{pmatrix} 8 & -19 \\ -\frac{29}{3} & 23 \end{pmatrix}$

$\mathbf{B}^{-1} = \frac{1}{3}\begin{pmatrix} 6 & -3 \\ -7 & 4 \end{pmatrix} = \begin{pmatrix} 2 & -1 \\ -\frac{7}{3} & \frac{4}{3} \end{pmatrix}$

$\mathbf{A}^{-1} = \frac{1}{1}\begin{pmatrix} 3 & -7 \\ -2 & 5 \end{pmatrix} = \begin{pmatrix} 3 & -7 \\ -2 & 5 \end{pmatrix}$

$\mathbf{B}^{-1}\mathbf{A}^{-1} = \begin{pmatrix} 2 & -1 \\ -\frac{7}{3} & \frac{4}{3} \end{pmatrix}\begin{pmatrix} 3 & -7 \\ -2 & 5 \end{pmatrix} = \begin{pmatrix} 8 & -19 \\ -\frac{29}{3} & 23 \end{pmatrix}$

$\therefore (\mathbf{AB})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1}$

**25) a)**

$\det \mathbf{A} = 8 \times 2 - 2 \times 7 = 2$

$\mathbf{A}^{-1} = \frac{1}{2}\begin{pmatrix} 2 & -2 \\ -7 & 8 \end{pmatrix} = \begin{pmatrix} 1 & -1 \\ -\frac{7}{2} & 4 \end{pmatrix}$

**25) b)**

$\det \mathbf{B} = (-1) \times 2 - (-1) \times 3 = 1$

$\mathbf{B}^{-1} = \frac{1}{1}\begin{pmatrix} 2 & 1 \\ -3 & -1 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ -3 & -1 \end{pmatrix}$

**25) c)**

$(\mathbf{AB})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1} = \begin{pmatrix} 2 & 1 \\ -3 & -1 \end{pmatrix}\begin{pmatrix} 1 & -1 \\ -\frac{7}{2} & 4 \end{pmatrix} = \begin{pmatrix} -\frac{3}{2} & 2 \\ \frac{1}{2} & -1 \end{pmatrix}$

**26) a)**

$\det \mathbf{A} = 3 \times 1 - 2 \times 2 = -1$

$\mathbf{A}^{-1} = \frac{1}{-1}\begin{pmatrix} 1 & -2 \\ -2 & 3 \end{pmatrix} = \begin{pmatrix} -1 & 2 \\ 2 & -3 \end{pmatrix}$

**26) b)**

$\det \mathbf{B} = 5 \times 2 - 3 \times 3 = 1$

$\mathbf{B}^{-1} = \frac{1}{1}\begin{pmatrix} 2 & -3 \\ -3 & 5 \end{pmatrix} = \begin{pmatrix} 2 & -3 \\ -3 & 5 \end{pmatrix}$

**26) c)**

$\det \mathbf{C} = 6 \times 3 - 7 \times 3 = -3$

$\mathbf{C}^{-1} = \frac{1}{-3}\begin{pmatrix} 3 & -7 \\ -3 & 6 \end{pmatrix} = \begin{pmatrix} -1 & \frac{7}{3} \\ 1 & -2 \end{pmatrix}$

**26) d)**

$(\mathbf{ABC})^{-1} = \mathbf{C}^{-1}\mathbf{B}^{-1}\mathbf{A}^{-1} = \begin{pmatrix} -1 & \frac{7}{3} \\ 1 & -2 \end{pmatrix}\begin{pmatrix} 2 & -3 \\ -3 & 5 \end{pmatrix}\begin{pmatrix} -1 & 2 \\ 2 & -3 \end{pmatrix} = \begin{pmatrix} \frac{115}{3} & -62 \\ -34 & 55 \end{pmatrix}$

**27) a)**

$\begin{pmatrix} k & -4 \\ 2 & 3 \end{pmatrix}^2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \Rightarrow \begin{pmatrix} k^2 - 8 & -4k - 12 \\ 2k + 6 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

$k^2 - 8 = 1 \Rightarrow k^2 = 9 \Rightarrow k = \pm 3$

$2k + 6 = 0 \Rightarrow k = -3$

$-4k - 12 = 0 \Rightarrow k = -3$

The matrix is self-inverse when $k = -3$

**27) b)**

$\begin{pmatrix} k & -35 \\ 1 & -k \end{pmatrix}^2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \Rightarrow \begin{pmatrix} k^2 - 35 & 0 \\ 0 & k^2 - 35 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

$k^2 - 35 = 1 \Rightarrow k^2 = 36 \Rightarrow k = \pm 6$

The matrix is self-inverse when $k = 6$ or $k = -6$

**27) c)**

$\begin{pmatrix} -3 & k \\ -2k & 3 \end{pmatrix}^2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \Rightarrow \begin{pmatrix} -2k^2 + 9 & 0 \\ 0 & -2k^2 + 9 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

$-2k^2 + 9 = 1 \Rightarrow k^2 = 4 \Rightarrow k = \pm 2$

The matrix is self-inverse when $k = 2$ or $k = -2$

**27) d)**

$\begin{pmatrix} k + 1 & 4 \\ -k & -3 \end{pmatrix}^2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \Rightarrow \begin{pmatrix} k^2 - 2k + 1 & 4k - 8 \\ -k^2 + 2k & -4k + 9 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

$k^2 - 2k + 1 = 1 \Rightarrow k^2 - 2k = 0 \Rightarrow k(k - 2) \Rightarrow k = 0$ or $k = 2$

$-k^2 + 2k = 0 \Rightarrow k^2 - 2k = 0 \Rightarrow k(k - 2) \Rightarrow k = 0$ or $k = 2$

$4k - 8 = 0 \Rightarrow k = 2$

$-4k + 9 = 1 \Rightarrow k = 2$

The matrix is self-inverse when $k = 2$

**27) e)**

$$\begin{pmatrix} -\frac{\sqrt{2}}{2} & 2k-1 \\ -4k & \frac{\sqrt{2}}{2} \end{pmatrix}^2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \Rightarrow \begin{pmatrix} \frac{-16k^2+8k+1}{2} & 0 \\ 0 & \frac{-16k^2+8k+1}{2} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$\frac{-16k^2+8k+1}{2} = 1 \Rightarrow -16k^2+8k+1 = 2 \Rightarrow -16k^2+8k-1 = 0$

$\Rightarrow 16k^2-8k+1 = 0 \Rightarrow (4k-1)^2 = 0 \Rightarrow k = \frac{1}{4}$

The matrix is self-inverse when $k = \frac{1}{4}$

**28)**

If $\mathbf{A}$ is self-inverse then $\mathbf{A}^{-1} = \mathbf{A}$

Pre-multiply by $\mathbf{A}$:

$\mathbf{A} \times \mathbf{A}^{-1} = \mathbf{A} \times \mathbf{A}$

$\Rightarrow \mathbf{I} = \mathbf{A}^2$

**29) a)**

$$\mathbf{AB} = k\mathbf{I} \Rightarrow \begin{pmatrix} a & 6 \\ b & c \end{pmatrix}\begin{pmatrix} -3 & 2 \\ -4 & 6 \end{pmatrix} = \begin{pmatrix} k & 0 \\ 0 & k \end{pmatrix} \Rightarrow \begin{pmatrix} -3a-24 & 2a+36 \\ -3b-4c & 2b+6c \end{pmatrix} = \begin{pmatrix} k & 0 \\ 0 & k \end{pmatrix}$$

$2a+36 = 0 \Rightarrow a = -18$

$-3a-24 = k \Rightarrow -3 \times (-18) - 24 = k \Rightarrow k = 30$

$-3b-4c = 0$ (1)

$2b+6c = 30$ (2)

$\therefore b = -12$ and $c = 9$

$\therefore a = -18$, $b = -12$, $c = 9$ and $k = 30$

**29) b)**

$\mathbf{AB} = 30\mathbf{I} \Rightarrow \mathbf{A}^{-1}\mathbf{AB} = \mathbf{A}^{-1} \times 30\mathbf{I} \Rightarrow \mathbf{IB} = 30\mathbf{A}^{-1} \Rightarrow \mathbf{A}^{-1} = \frac{1}{30}\mathbf{B}$

$$\therefore \mathbf{A}^{-1} = \frac{1}{30}\begin{pmatrix} -3 & 2 \\ -4 & 6 \end{pmatrix} = \begin{pmatrix} -\frac{1}{10} & \frac{1}{15} \\ -\frac{2}{15} & \frac{1}{5} \end{pmatrix}$$

**30) a)**

$$(\mathbf{BA})^{-1} = \mathbf{A}^{-1}\mathbf{B}^{-1} = \begin{pmatrix} \frac{5}{6} & -\frac{1}{3} \\ -\frac{2}{3} & \frac{2}{3} \end{pmatrix}$$

$$\begin{vmatrix} \frac{5}{6} & -\frac{1}{3} \\ -\frac{2}{3} & \frac{2}{3} \end{vmatrix} = \frac{5}{6} \times \frac{2}{3} - \left(-\frac{1}{3}\right) \times \left(-\frac{2}{3}\right) = \frac{1}{3}$$

$$\begin{pmatrix} \frac{5}{6} & -\frac{1}{3} \\ -\frac{2}{3} & \frac{2}{3} \end{pmatrix}^{-1} = \frac{1}{\frac{1}{3}} \begin{pmatrix} \frac{2}{3} & \frac{1}{3} \\ \frac{2}{3} & \frac{5}{6} \end{pmatrix} = 3 \begin{pmatrix} \frac{2}{3} & \frac{1}{3} \\ \frac{2}{3} & \frac{5}{6} \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 2 & \frac{5}{2} \end{pmatrix}$$

$$\therefore \mathbf{BA} = \begin{pmatrix} 2 & 1 \\ 2 & \frac{5}{2} \end{pmatrix}$$

**30) b)**

$$\mathbf{BA} = \begin{pmatrix} 2 & 1 \\ 2 & \frac{5}{2} \end{pmatrix} \Rightarrow \mathbf{B}^{-1}\mathbf{BA} = \mathbf{B}^{-1} \begin{pmatrix} 2 & 1 \\ 2 & \frac{5}{2} \end{pmatrix} \Rightarrow \mathbf{IA} = \mathbf{B}^{-1} \begin{pmatrix} 2 & 1 \\ 2 & \frac{5}{2} \end{pmatrix} \Rightarrow \mathbf{A} = \mathbf{B}^{-1} \begin{pmatrix} 2 & 1 \\ 2 & \frac{5}{2} \end{pmatrix}$$

$$\det \mathbf{B} = 1 \times \frac{3}{2} - 0 \times 1 = \frac{3}{2}$$

$$\mathbf{B}^{-1} = \frac{1}{\frac{3}{2}} \begin{pmatrix} \frac{3}{2} & 0 \\ -1 & 1 \end{pmatrix} = \frac{2}{3} \begin{pmatrix} \frac{3}{2} & 0 \\ -1 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ -\frac{2}{3} & \frac{2}{3} \end{pmatrix}$$

$$\therefore \mathbf{A} = \begin{pmatrix} 1 & 0 \\ -\frac{2}{3} & \frac{2}{3} \end{pmatrix} \begin{pmatrix} 2 & 1 \\ 2 & \frac{5}{2} \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix}$$

**31) a)**

$$\det(\mathbf{A}^{-1}) = \frac{1}{\det \mathbf{A}} = \frac{1}{5}$$

**31) b)**

$$\det(\mathbf{B}^{-1}) = \frac{1}{\det \mathbf{B}} = \frac{1}{\frac{2}{3}} = \frac{3}{2}$$

**31) c)**

$$\det((\mathbf{AB})^{-1}) = \det(\mathbf{B}^{-1}\mathbf{A}^{-1}) = \det(\mathbf{B}^{-1}) \times \det(\mathbf{A}^{-1}) = \frac{3}{2} \times \frac{1}{5} = \frac{3}{10}$$

**32) a)**

$$\det(\mathbf{A}^{-1}) = 4 \Rightarrow \det \mathbf{A} = \frac{1}{4}$$

$$\det(\mathbf{A}^4) = (\det \mathbf{A})^4 = \left(\frac{1}{4}\right)^4 = \frac{1}{256}$$

**32) b)**

$\det(\mathbf{AB}) = \det\mathbf{A} \times \det\mathbf{B} = \frac{1}{4} \times -2 = -\frac{1}{2}$

**33)**

$\det(\mathbf{A}^{-1}) = -\frac{1}{5} \Rightarrow \det\mathbf{A} = -5$

$\det(\mathbf{AB}) = 40 \Rightarrow \det\mathbf{A} \times \det\mathbf{B} = 40 \Rightarrow -5 \times \det\mathbf{B} = 40 \Rightarrow \det\mathbf{B} = -8$

$\det(\mathbf{B}^5) = (\det\mathbf{B})^5 = (-8)^5 = -32768$

**34) a)**

$$\mathbf{AB} = \begin{pmatrix} 10 & -6 & 2 \\ 3 & -2 & 1 \\ -4 & 4 & -2 \end{pmatrix}\begin{pmatrix} 0 & 2 & 1 \\ -1 & 6 & 2 \\ -2 & 8 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix}$$

**34) b)**

$\mathbf{AB} = 2\mathbf{I} \Rightarrow \mathbf{A}^{-1}\mathbf{AB} = \mathbf{A}^{-1} \times 2\mathbf{I} \Rightarrow \mathbf{IB} = 2\mathbf{A}^{-1} \Rightarrow \frac{1}{2}\mathbf{B} = \mathbf{A}^{-1}$

$$\therefore \mathbf{A}^{-1} = \frac{1}{2}\begin{pmatrix} 0 & 2 & 1 \\ -1 & 6 & 2 \\ -2 & 8 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 1 & \frac{1}{2} \\ -\frac{1}{2} & 3 & 1 \\ -1 & 4 & \frac{1}{2} \end{pmatrix}$$

**35) a)**

$$\mathbf{AB} = k\mathbf{I} \Rightarrow \begin{pmatrix} p & 0 & q \\ 2 & 1 & 4 \\ -2 & 0 & r \end{pmatrix}\begin{pmatrix} 0 & 0 & -2 \\ 4 & 4 & 1 \\ -1 & 0 & \frac{3}{4} \end{pmatrix} = \begin{pmatrix} k & 0 & 0 \\ 0 & k & 0 \\ 0 & 0 & k \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} -q & 0 & \frac{3q}{4} - 2p \\ 0 & 4 & 0 \\ -r & 0 & \frac{3r}{4} + 4 \end{pmatrix} = \begin{pmatrix} k & 0 & 0 \\ 0 & k & 0 \\ 0 & 0 & k \end{pmatrix}$$

$k = 4$

$-q = k \Rightarrow -q = 4 \Rightarrow q = -4$

$-r = 0 \Rightarrow r = 0$

$\frac{3q}{4} - 2p = 0 \Rightarrow \frac{3(-4)}{4} - 2p = 0 \Rightarrow p = -\frac{3}{2}$

**35) b)**

$\mathbf{AB} = 4\mathbf{I} \Rightarrow \mathbf{A}^{-1}\mathbf{AB} = \mathbf{A}^{-1} \times 4\mathbf{I} \Rightarrow \mathbf{IB} = 4\mathbf{A}^{-1} \Rightarrow \mathbf{A}^{-1} = \frac{1}{4}\mathbf{B}$

$$\therefore \mathbf{A}^{-1} = \frac{1}{4}\begin{pmatrix} 0 & 0 & -2 \\ 4 & 4 & 1 \\ -1 & 0 & \frac{3}{4} \end{pmatrix} = \begin{pmatrix} 0 & 0 & -\frac{1}{2} \\ 1 & 1 & \frac{1}{4} \\ -\frac{1}{4} & 0 & \frac{3}{16} \end{pmatrix}$$

**36) a)**

*Calculator:*

$$\begin{pmatrix} 2 & 1 & 0 \\ 7 & 1 & 2 \\ -3 & -3 & 1 \end{pmatrix}^{-1} = \begin{pmatrix} 7 & -1 & 2 \\ -13 & 2 & -4 \\ -18 & 3 & -5 \end{pmatrix}$$

*By hand:*

Find the cofactor matrix:

$$\begin{pmatrix} 7 & -13 & -18 \\ -1 & 2 & 3 \\ 2 & -4 & -5 \end{pmatrix}$$

Find the transpose of the cofactor matrix:

$$\begin{pmatrix} 7 & -1 & 2 \\ -13 & 2 & -4 \\ -18 & 3 & -5 \end{pmatrix}$$

Find the determinant of the original matrix:

$$\begin{vmatrix} 2 & 1 & 0 \\ 7 & 1 & 2 \\ -3 & -3 & 1 \end{vmatrix} = 2\begin{vmatrix} 1 & 2 \\ -3 & 1 \end{vmatrix} - 1\begin{vmatrix} 7 & 2 \\ -3 & 1 \end{vmatrix} + 0\begin{vmatrix} 7 & 1 \\ -3 & -3 \end{vmatrix}$$

$$= 2\big(1 \times 1 - 2 \times (-3)\big) - \big(7 \times 1 - 2 \times (-3)\big) + 0 = 2 \times 7 - 13 = 1$$

Then:

$$\begin{pmatrix} 2 & 1 & 0 \\ 7 & 1 & 2 \\ -3 & -3 & 1 \end{pmatrix}^{-1} = \frac{1}{1}\begin{pmatrix} 7 & -1 & 2 \\ -13 & 2 & -4 \\ -18 & 3 & -5 \end{pmatrix} = \begin{pmatrix} 7 & -1 & 2 \\ -13 & 2 & -4 \\ -18 & 3 & -5 \end{pmatrix}$$

**36) b)**

*Calculator:*

$$\begin{pmatrix} 2 & 1 & -2 \\ 0 & 0 & 1 \\ 4 & 3 & -1 \end{pmatrix}^{-1} = \begin{pmatrix} \frac{3}{2} & \frac{5}{2} & -\frac{1}{2} \\ -2 & -3 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

*By hand:*

Find the cofactor matrix:

$$\begin{pmatrix} -3 & 4 & 0 \\ -5 & 6 & -2 \\ 1 & -2 & 0 \end{pmatrix}$$

Find the transpose of the cofactor matrix:

$$\begin{pmatrix} -3 & -5 & 1 \\ 4 & 6 & -2 \\ 0 & -2 & 0 \end{pmatrix}$$

Find the determinant of the original matrix:

$$\begin{vmatrix} 2 & 1 & -2 \\ 0 & 0 & 1 \\ 4 & 3 & -1 \end{vmatrix} = 2\begin{vmatrix} 0 & 1 \\ 3 & -1 \end{vmatrix} - 1\begin{vmatrix} 0 & 1 \\ 4 & -1 \end{vmatrix} - 2\begin{vmatrix} 0 & 0 \\ 4 & 3 \end{vmatrix}$$

$$= 2(0 \times (-1) - 1 \times 3) - (0 \times (-1) - 1 \times 4) - 2(0 \times 3 - 0 \times 4)$$

$$= 2 \times (-3) - (-4) - 0 = -2$$

Then:

$$\begin{pmatrix} 2 & 1 & -2 \\ 0 & 0 & 1 \\ 4 & 3 & -1 \end{pmatrix}^{-1} = \frac{1}{-2}\begin{pmatrix} -3 & -5 & 1 \\ 4 & 6 & -2 \\ 0 & -2 & 0 \end{pmatrix} = \begin{pmatrix} \frac{3}{2} & \frac{5}{2} & -\frac{1}{2} \\ -2 & -3 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

**36) c)**

*Calculator:*

$$\begin{pmatrix}4 & 0 & 3\\4 & 3 & 6\\5 & -1 & 3\end{pmatrix}^{-1} = \begin{pmatrix}5 & -1 & -3\\6 & -1 & -4\\-\frac{19}{3} & \frac{4}{3} & 4\end{pmatrix}$$

*By hand:*

Find the cofactor matrix:

$$\begin{pmatrix}15 & 18 & -19\\-3 & -3 & 4\\-9 & -12 & 12\end{pmatrix}$$

Find the transpose of the cofactor matrix:

$$\begin{pmatrix}15 & -3 & -9\\18 & -3 & -12\\-19 & 4 & 12\end{pmatrix}$$

Find the determinant of the original matrix:

$$\begin{vmatrix}4 & 0 & 3\\4 & 3 & 6\\5 & -1 & 3\end{vmatrix} = 4\begin{vmatrix}3 & 6\\-1 & 3\end{vmatrix} - 0\begin{vmatrix}4 & 6\\5 & 3\end{vmatrix} + 3\begin{vmatrix}4 & 3\\5 & -1\end{vmatrix}$$

$$= 4\big(3\times 3 - 6\times(-1)\big) - 0(4\times 3 - 6\times 5) + 3(4\times(-1) - 3\times 5)$$

$$= 4\times 15 - 0 + 3\times -19 = 3$$

Then:

$$\begin{pmatrix}4 & 0 & 3\\4 & 3 & 6\\5 & -1 & 3\end{pmatrix}^{-1} = \frac{1}{3}\begin{pmatrix}15 & -3 & -9\\18 & -3 & -12\\-19 & 4 & 12\end{pmatrix} = \begin{pmatrix}5 & -1 & -3\\6 & -1 & -4\\-\frac{19}{3} & \frac{4}{3} & 4\end{pmatrix}$$

**37) a)**

*Calculator:*

$$\begin{pmatrix}1 & 1 & 2\\ 1 & 1 & -3\\ 1 & 2 & -3\end{pmatrix}^{-1} = \begin{pmatrix}\frac{3}{5} & \frac{7}{5} & -1\\ 0 & -1 & 1\\ \frac{1}{5} & -\frac{1}{5} & 0\end{pmatrix}$$

*By hand:*

Find the cofactor matrix:

$$\begin{pmatrix}3 & 0 & 1\\ 7 & -5 & -1\\ -5 & 5 & 0\end{pmatrix}$$

Find the transpose of the cofactor matrix:

$$\begin{pmatrix}3 & 7 & -5\\ 0 & -5 & 5\\ 1 & -1 & 0\end{pmatrix}$$

Find the determinant of the original matrix:

$$\begin{vmatrix}1 & 1 & 2\\ 1 & 1 & -3\\ 1 & 2 & -3\end{vmatrix} = 1\begin{vmatrix}1 & -3\\ 2 & -3\end{vmatrix} - 1\begin{vmatrix}1 & -3\\ 1 & -3\end{vmatrix} + 2\begin{vmatrix}1 & 1\\ 1 & 2\end{vmatrix}$$

$$= 1(1\times(-3) - (-3)\times 2) - 1(1\times(-3) - (-3)\times 1) + 2(1\times 2 - 1\times 1)$$

$$= 1\times 3 - 1\times 0 + 2\times 1 = 5$$

Then:

$$\begin{pmatrix}1 & 1 & 2\\ 1 & 1 & -3\\ 1 & 2 & -3\end{pmatrix}^{-1} = \frac{1}{5}\begin{pmatrix}3 & 7 & -5\\ 0 & -5 & 5\\ 1 & -1 & 0\end{pmatrix} = \begin{pmatrix}\frac{3}{5} & \frac{7}{5} & -1\\ 0 & -1 & 1\\ \frac{1}{5} & -\frac{1}{5} & 0\end{pmatrix}$$

**37) b)**

Let $P$ have coordinates $(a, b, c)$

$$\mathbf{A}\begin{pmatrix}a\\ b\\ c\end{pmatrix} = \begin{pmatrix}1\\ 3\\ -2\end{pmatrix} \Rightarrow \mathbf{A}^{-1}\mathbf{A}\begin{pmatrix}a\\ b\\ c\end{pmatrix} = \mathbf{A}^{-1}\begin{pmatrix}1\\ 3\\ -2\end{pmatrix} \Rightarrow \mathbf{I}\begin{pmatrix}a\\ b\\ c\end{pmatrix} = \mathbf{A}^{-1}\begin{pmatrix}1\\ 3\\ -2\end{pmatrix}$$

$$\Rightarrow \begin{pmatrix}a\\ b\\ c\end{pmatrix} = \begin{pmatrix}\frac{3}{5} & \frac{7}{5} & -1\\ 0 & -1 & 1\\ \frac{1}{5} & -\frac{1}{5} & 0\end{pmatrix}\begin{pmatrix}1\\ 3\\ -2\end{pmatrix} = \begin{pmatrix}\frac{34}{5}\\ -5\\ -\frac{2}{5}\end{pmatrix}$$

So $P$ has coordinates $\left(\frac{34}{5}, -5, -\frac{2}{5}\right)$

**38)**

First find $\mathbf{B}^{-1}$:

*Calculator:*

$$\begin{pmatrix} 6 & -10 & -10 \\ 0 & -3 & -3 \\ 3 & -5 & -\frac{19}{2} \end{pmatrix}^{-1} = \begin{pmatrix} \frac{1}{6} & -\frac{5}{9} & 0 \\ -\frac{1}{9} & -\frac{1}{3} & \frac{2}{9} \\ \frac{1}{9} & 0 & -\frac{2}{9} \end{pmatrix}$$

*By hand:*

Find the cofactor matrix:

$$\begin{pmatrix} \frac{27}{2} & -9 & 9 \\ -45 & -27 & 0 \\ 0 & 18 & -18 \end{pmatrix}$$

Find the transpose of the cofactor matrix:

$$\begin{pmatrix} \frac{27}{2} & -45 & 0 \\ -9 & -27 & 18 \\ 9 & 0 & -18 \end{pmatrix}$$

Find the determinant of the original matrix:

$$\begin{vmatrix} 6 & -10 & -10 \\ 0 & -3 & -3 \\ 3 & -5 & -\frac{19}{2} \end{vmatrix} = 6\begin{vmatrix} -3 & -3 \\ -5 & -\frac{19}{2} \end{vmatrix} + 10\begin{vmatrix} 0 & -3 \\ 3 & -\frac{19}{2} \end{vmatrix} - 10\begin{vmatrix} 0 & -3 \\ 3 & -5 \end{vmatrix}$$

$$= 6\left((-3) \times \left(-\frac{19}{2}\right) - (-3) \times (-5)\right) + 10\left(0 \times \left(-\frac{19}{2}\right) - (-3) \times 3\right) - 10(0 \times (-5) - (-3) \times 3)$$

$$= 6 \times \frac{27}{2} + 10 \times 9 - 10 \times 9 = 81$$

Then:

$$\begin{pmatrix} 6 & -10 & -10 \\ 0 & -3 & -3 \\ 3 & -5 & -\frac{19}{2} \end{pmatrix}^{-1} = \frac{1}{81}\begin{pmatrix} \frac{27}{2} & -45 & 0 \\ -9 & -27 & 18 \\ 9 & 0 & -18 \end{pmatrix} = \begin{pmatrix} \frac{1}{6} & -\frac{5}{9} & 0 \\ -\frac{1}{9} & -\frac{1}{3} & \frac{2}{9} \\ \frac{1}{9} & 0 & -\frac{2}{9} \end{pmatrix}$$

So:

$$-\frac{1}{9}\begin{pmatrix} -\frac{3}{2} & 5 & 0 \\ 1 & 3 & -2 \\ -1 & 0 & 2 \end{pmatrix} = \begin{pmatrix} \frac{1}{6} & -\frac{5}{9} & 0 \\ -\frac{1}{9} & -\frac{1}{3} & \frac{2}{9} \\ \frac{1}{9} & 0 & -\frac{2}{9} \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} -\frac{3}{2} & 5 & 0 \\ 1 & 3 & -2 \\ -1 & 0 & 2 \end{pmatrix} = -9 \begin{pmatrix} \frac{1}{6} & -\frac{5}{9} & 0 \\ -\frac{1}{9} & -\frac{1}{3} & \frac{2}{9} \\ \frac{1}{9} & 0 & -\frac{2}{9} \end{pmatrix} \Rightarrow \mathbf{A} = -9\mathbf{B}^{-1}$$

$\therefore k = -9$

**39) a)**

Find the cofactor matrix:

$$\begin{pmatrix} -1 & 1 & 0 \\ 1 & -1 & 1-k \\ 0 & 1-k & k-1 \end{pmatrix}$$

Find the transpose of the cofactor matrix:

$$\begin{pmatrix} -1 & 1 & 0 \\ 1 & -1 & 1-k \\ 0 & 1-k & k-1 \end{pmatrix}$$

Find the determinant of the original matrix:

$$\begin{vmatrix} k & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 0 \end{vmatrix} = k \begin{vmatrix} 1 & 1 \\ 1 & 0 \end{vmatrix} - 1 \begin{vmatrix} 1 & 1 \\ 1 & 0 \end{vmatrix} + 1 \begin{vmatrix} 1 & 1 \\ 1 & 1 \end{vmatrix}$$

$= k(1 \times 0 - 1 \times 1) - 1(1 \times 0 - 1 \times 1) + 1(1 \times 1 - 1 \times 1)$

$= k \times (-1) - 1 \times (-1) + 1 \times 0 = 1 - k$

Then:

$$\begin{pmatrix} k & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 0 \end{pmatrix}^{-1} = \frac{1}{1-k} \begin{pmatrix} -1 & 1 & 0 \\ 1 & -1 & 1-k \\ 0 & 1-k & k-1 \end{pmatrix} = \begin{pmatrix} -\frac{1}{1-k} & \frac{1}{1-k} & 0 \\ \frac{1}{1-k} & -\frac{1}{1-k} & 1 \\ 0 & 1 & -1 \end{pmatrix}$$

**39) b)**

Find the cofactor matrix:

$$\begin{pmatrix} 4 & -12 & 1 \\ k & 8-2k & -2 \\ -k & 3k & 2 \end{pmatrix}$$

Find the transpose of the cofactor matrix:

$$\begin{pmatrix} 4 & k & -k \\ -12 & 8-2k & 3k \\ 1 & -2 & 2 \end{pmatrix}$$

Find the determinant of the original matrix:

$$\begin{vmatrix} 2 & 0 & k \\ 3 & 1 & 0 \\ 2 & 1 & 4 \end{vmatrix} = 2\begin{vmatrix} 1 & 0 \\ 1 & 4 \end{vmatrix} - 0\begin{vmatrix} 3 & 0 \\ 2 & 4 \end{vmatrix} + k\begin{vmatrix} 3 & 1 \\ 2 & 1 \end{vmatrix}$$

$$= 2(1 \times 4 - 0 \times 1) - 0(3 \times 4 - 0 \times 2) + k(3 \times 1 - 1 \times 2)$$

$$= 2 \times 4 - 0 + k \times 1 = k + 8$$

Then:

$$\begin{pmatrix} 2 & 0 & k \\ 3 & 1 & 0 \\ 2 & 1 & 4 \end{pmatrix}^{-1} = \frac{1}{k+8}\begin{pmatrix} 4 & k & -k \\ -12 & 8-2k & 3k \\ 1 & -2 & 2 \end{pmatrix} = \begin{pmatrix} \frac{4}{k+8} & \frac{k}{k+8} & -\frac{k}{k+8} \\ -\frac{12}{k+8} & \frac{8-2k}{k+8} & \frac{3k}{k+8} \\ \frac{1}{k+8} & -\frac{2}{k+8} & \frac{2}{k+8} \end{pmatrix}$$

**39) c)**

Find the cofactor matrix:

$$\begin{pmatrix} -18 & -3k & 3k \\ -2 & 5k & 3k \\ 14 & k & -9k \end{pmatrix}$$

Find the transpose of the cofactor matrix:

$$\begin{pmatrix} -18 & -2 & 14 \\ -3k & 5k & k \\ 3k & 3k & -9k \end{pmatrix}$$

Find the determinant of the original matrix:

$$\begin{vmatrix} 2k & -1 & 3 \\ k & -5 & 1 \\ k & -2 & 4 \end{vmatrix} = 2k\begin{vmatrix} -5 & 1 \\ -2 & 4 \end{vmatrix} + 1\begin{vmatrix} k & 1 \\ k & 4 \end{vmatrix} + 3\begin{vmatrix} k & -5 \\ k & -2 \end{vmatrix}$$

$$= 2k\big(-5 \times 4 - 1 \times (-2)\big) + 1(k \times 4 - 1 \times k) + 3(k \times (-2) - (-5) \times k)$$

$$= 2k \times (-18) + 1 \times 3k + 3 \times 3k = -24k$$

Then:

$$\begin{pmatrix} 2k & -1 & 3 \\ k & -5 & 1 \\ k & -2 & 4 \end{pmatrix}^{-1} = \frac{1}{-24k}\begin{pmatrix} -18 & -2 & 14 \\ -3k & 5k & k \\ 3k & 3k & -9k \end{pmatrix} = \begin{pmatrix} \frac{3}{4k} & \frac{1}{12k} & -\frac{7}{12k} \\ \frac{1}{8} & -\frac{5}{24} & -\frac{1}{24} \\ -\frac{1}{8} & -\frac{1}{8} & \frac{3}{8} \end{pmatrix}$$

**39) d)**

Find the cofactor matrix:

$$\begin{pmatrix} 3-2k & -2k & 2k+3 \\ -4 & -2 & 0 \\ 2k+3 & 0 & -2k-3 \end{pmatrix}$$

Find the transpose of the cofactor matrix:

$$\begin{pmatrix} 3-2k & -4 & 2k+3 \\ -2k & -2 & 0 \\ 2k+3 & 0 & -2k-3 \end{pmatrix}$$

Find the determinant of the original matrix:

$$\begin{vmatrix} -1 & 2 & -1 \\ k & 3 & k \\ -1 & 2 & 1 \end{vmatrix} = -1\begin{vmatrix} 3 & k \\ 2 & 1 \end{vmatrix} - 2\begin{vmatrix} k & k \\ -1 & 1 \end{vmatrix} - 1\begin{vmatrix} k & 3 \\ -1 & 2 \end{vmatrix}$$

$$= -1(3 \times 1 - k \times 2) - 2\big(k \times 1 - k \times (-1)\big) - 1\big(k \times 2 - 3 \times (-1)\big)$$

$$= -1 \times (3-2k) - 2 \times 2k - 1 \times (2k+3) = -4k - 6$$

Then:

$$\begin{pmatrix} -1 & 2 & -1 \\ k & 3 & k \\ -1 & 2 & 1 \end{pmatrix}^{-1} = \frac{1}{-4k-6}\begin{pmatrix} 3-2k & -4 & 2k+3 \\ -2k & -2 & 0 \\ 2k+3 & 0 & -2k-3 \end{pmatrix} = \begin{pmatrix} \frac{3-2k}{-4k-6} & -\frac{4}{-4k-6} & \frac{2k+3}{-4k-6} \\ -\frac{2k}{-4k-6} & -\frac{2}{-4k-6} & 0 \\ \frac{2k+3}{-4k-6} & 0 & \frac{-2k-3}{-4k-6} \end{pmatrix}$$

$$= \begin{pmatrix} \frac{2k-3}{4k+6} & \frac{2}{2k+3} & -\frac{1}{2} \\ \frac{k}{2k+3} & \frac{1}{2k+3} & 0 \\ -\frac{1}{2} & 0 & \frac{1}{2} \end{pmatrix}$$

**40) a)**

$$\begin{vmatrix} 1 & -1 & k \\ k & 9 & 18 \\ 3 & 0 & 27 \end{vmatrix} = 1\begin{vmatrix} 9 & 18 \\ 0 & 27 \end{vmatrix} + 1\begin{vmatrix} k & 18 \\ 3 & 27 \end{vmatrix} + k\begin{vmatrix} k & 9 \\ 3 & 0 \end{vmatrix}$$

$$= 1(9 \times 27 - 18 \times 0) + 1(k \times 27 - 18 \times 3) + k(k \times 0 - 9 \times 3)$$

$$= 1 \times 243 + 1 \times (27k - 54) + k \times (-27) = 243 + 27k - 54 - 27k = 189$$

So $\mathbf{A}$ is a non-singular matrix for all values of $k$

**40) b)**

Find the cofactor matrix:

$$\begin{pmatrix} 243 & 54-27k & -27 \\ 27 & 27-3k & -3 \\ -9k-18 & k^2-18 & k+9 \end{pmatrix}$$

Find the transpose of the cofactor matrix:

$$\begin{pmatrix} 243 & 27 & -9k-18 \\ 54-27k & 27-3k & k^2-18 \\ -27 & -3 & k+9 \end{pmatrix}$$

Then:

$$\begin{pmatrix} 1 & -1 & k \\ k & 9 & 18 \\ 3 & 0 & 27 \end{pmatrix}^{-1} = \frac{1}{189}\begin{pmatrix} 243 & 27 & -9k-18 \\ 54-27k & 27-3k & k^2-18 \\ -27 & -3 & k+9 \end{pmatrix} = \begin{pmatrix} \frac{9}{7} & \frac{1}{7} & \frac{-9k-18}{189} \\ \frac{54-27k}{189} & \frac{27-3k}{189} & \frac{k^2-18}{189} \\ -\frac{1}{7} & -\frac{1}{63} & \frac{k+9}{189} \end{pmatrix}$$

$$= \begin{pmatrix} \frac{9}{7} & \frac{1}{7} & -\frac{k+2}{21} \\ \frac{2-k}{7} & \frac{9-k}{63} & \frac{k^2-18}{189} \\ -\frac{1}{7} & -\frac{1}{63} & \frac{k+9}{189} \end{pmatrix}$$

**41)**

Find the cofactor matrix:

$$\begin{pmatrix} 5-5k & 5k-5 & k^2+k-2 \\ 18-2k & 5k-27 & k^2-4k-9 \\ 16 & 3k-25 & -2k-10 \end{pmatrix}$$

Find the transpose of the cofactor matrix:

$$\begin{pmatrix} 5-5k & 18-2k & 16 \\ 5k-5 & 5k-27 & 3k-25 \\ k^2+k-2 & k^2-4k-9 & -2k-10 \end{pmatrix}$$

Find the determinant of the original matrix:

$$\begin{vmatrix} 5-k & 4 & -2 \\ k & -2 & 5 \\ -1 & k+1 & -5 \end{vmatrix} = (5-k)\begin{vmatrix} -2 & 5 \\ k+1 & -5 \end{vmatrix} - 4\begin{vmatrix} k & 5 \\ -1 & -5 \end{vmatrix} - 2\begin{vmatrix} k & -2 \\ -1 & k+1 \end{vmatrix}$$

$$= (5-k)\big(-2\times(-5) - 5\times(k+1)\big) - 4\big(k\times(-5) - 5\times(-1)\big) - 2\big(k\times(k+1) - (-2)\times(-1)\big)$$

$$= (5-k)\times(5-5k) - 4\times(5-5k) - 2\times(k^2+k-2)$$

$$= 5(5-k)(1-k) - 20(1-k) - 2(k+2)(k-1)$$

$$= 5(k-5)(k-1) + 20(k-1) - 2(k+2)(k-1)$$

$$= (k-1)\big(5(k-5) + 20 - 2(k+2)\big)$$

$$= (k-1)(3k-9) = 3(k-1)(k-3)$$

Then:

$$\begin{pmatrix} 5-k & 4 & -2 \\ k & -2 & 5 \\ -1 & k+1 & -5 \end{pmatrix}^{-1} = \frac{1}{3(k-1)(k-3)}\begin{pmatrix} 5-5k & 18-2k & 16 \\ 5k-5 & 5k-27 & 3k-25 \\ k^2+k-2 & k^2-4k-9 & -2k-10 \end{pmatrix}$$

$$= \frac{1}{3(k-1)(k-3)}\begin{pmatrix} -5(k-1) & -2(k-9) & 16 \\ 5(k-1) & 5k-27 & 3k-25 \\ (k+2)(k-1) & k^2-4k-9 & -2(k+5) \end{pmatrix}$$

$$= \begin{pmatrix} -\frac{5}{3(k-3)} & -\frac{2(k-9)}{3(k-1)(k-3)} & \frac{16}{3(k-1)(k-3)} \\ \frac{5}{3(k-3)} & \frac{5k-27}{3(k-1)(k-3)} & \frac{3k-25}{3(k-1)(k-3)} \\ \frac{k+2}{3(k-3)} & \frac{k^2-4k-9}{3(k-1)(k-3)} & -\frac{2(k+5)}{3(k-1)(k-3)} \end{pmatrix}$$

**A** is non-singular if $k \neq 1$ or $k \neq 3$

**42)**

Find the cofactor matrix:

$$\begin{pmatrix} -k-5 & 1 & -1 \\ 3k+7 & -k^2-5k-5 & -2k-1 \\ -3k-3 & k^2+4k+3 & 3k+3 \end{pmatrix}$$

Find the transpose of the cofactor matrix:

$$\begin{pmatrix} -k-5 & 3k+7 & -3k-3 \\ 1 & -k^2-5k-5 & k^2+4k+3 \\ -1 & -2k-1 & 3k+3 \end{pmatrix}$$

Find the determinant of the original matrix:

$$\begin{vmatrix} -2-k & -3 & 1 \\ -1 & -3 & k+2 \\ -1 & -2 & k+3 \end{vmatrix} = (-2-k)\begin{vmatrix} -3 & k+2 \\ -2 & k+3 \end{vmatrix} + 3\begin{vmatrix} -1 & k+2 \\ -1 & k+3 \end{vmatrix} + 1\begin{vmatrix} -1 & -3 \\ -1 & -2 \end{vmatrix}$$

$$= (-2-k)\big((-3)\times(k+3)-(k+2)\times(-2)\big) + 3\big((-1)\times(k+3)-(k+2)\times(-1)\big) + 1\big((-1)\times(-2)-(-3)\times(-1)\big)$$

$$= (-2-k)\times(-k-5) + 3\times(-1) + 1\times(-1) = k^2+7k+6 = (k+1)(k+6)$$

Then:

$$\begin{pmatrix} -2-k & -3 & 1 \\ -1 & -3 & k+2 \\ -1 & -2 & k+3 \end{pmatrix}^{-1} = \frac{1}{(k+1)(k+6)}\begin{pmatrix} -k-5 & 3k+7 & -3k-3 \\ 1 & -k^2-5k-5 & k^2+4k+3 \\ -1 & -2k-1 & 3k+3 \end{pmatrix}$$

$$= \frac{1}{(k+1)(k+6)}\begin{pmatrix} -(k+5) & 3k+7 & -3(k+1) \\ 1 & -(k^2+5k+5) & (k+1)(k+3) \\ -1 & -(2k+1) & 3(k+1) \end{pmatrix}$$

$$= \begin{pmatrix} -\frac{k+5}{(k+1)(k+6)} & \frac{3k+7}{(k+1)(k+6)} & -\frac{3}{k+6} \\ \frac{1}{(k+1)(k+6)} & -\frac{k^2+5k+5}{(k+1)(k+6)} & \frac{k+3}{k+6} \\ -\frac{1}{(k+1)(k+6)} & -\frac{2k+1}{(k+1)(k+6)} & \frac{3}{k+6} \end{pmatrix}$$ where $k \neq -1$ and $k \neq -6$

**43)**

Find the cofactor matrix:

$$\begin{pmatrix} 2k-9 & 0 & 0 \\ k^2-24 & 3 & -k \\ 3k-16 & 2 & -3 \end{pmatrix}$$

Find the transpose of the cofactor matrix:

$$\begin{pmatrix} 2k-9 & k^2-24 & 3k-16 \\ 0 & 3 & 2 \\ 0 & -k & -3 \end{pmatrix}$$

Find the determinant of the original matrix:

$$\begin{vmatrix} 1 & 8 & k \\ 0 & -3 & -2 \\ 0 & k & 3 \end{vmatrix} = 1\begin{vmatrix} -3 & -2 \\ k & 3 \end{vmatrix} - 8\begin{vmatrix} 0 & -2 \\ 0 & 3 \end{vmatrix} + k\begin{vmatrix} 0 & -3 \\ 0 & k \end{vmatrix}$$

$$= 1\big((-3)\times 3-(-2)\times k\big) - 8(0\times 3-(-2)\times 0) + k(0\times k-(-3)\times 0)$$

$$= 2k-9$$

Then:

$$\begin{pmatrix} 1 & 8 & k \\ 0 & -3 & -2 \\ 0 & k & 3 \end{pmatrix}^{-1} = \frac{1}{2k-9}\begin{pmatrix} 2k-9 & k^2-24 & 3k-16 \\ 0 & 3 & 2 \\ 0 & -k & -3 \end{pmatrix} = \begin{pmatrix} 1 & \frac{k^2-24}{2k-9} & \frac{3k-16}{2k-9} \\ 0 & \frac{3}{2k-9} & \frac{2}{2k-9} \\ 0 & -\frac{k}{2k-9} & -\frac{3}{2k-9} \end{pmatrix}$$

$\frac{k^2-24}{2k-9} = 8 \Rightarrow k^2-24 = 16k-72 \Rightarrow k^2-16k+48 = 0 \Rightarrow k = 4$ or $k = 12$

$\frac{3k-16}{2k-9} = k \Rightarrow 3k-16 = 2k^2-9k \Rightarrow 2k^2-12k+16 = 0 \Rightarrow k = 2$ or $k = 4$

$\frac{3}{2k-9} = -3 \Rightarrow 3 = -6k+27 \Rightarrow 6k = 24 \Rightarrow k = 4$

$\frac{2}{2k-9} = -2 \Rightarrow 2 = -4k+18 \Rightarrow 4k = 16 \Rightarrow k = 4$

$-\frac{k}{2k-9} = k \Rightarrow -k = 2k^2-9k \Rightarrow 2k^2-8k = 0 \Rightarrow k = 0$ or $k = 4$

$-\frac{3}{2k-9} = 3 \Rightarrow -3 = 6k-27 \Rightarrow 6k = 24 \Rightarrow k = 4$

$\therefore k = 4$

**44) a)**

$\begin{pmatrix}2 & 3\\ 1 & 1\end{pmatrix}\begin{pmatrix}x\\ y\end{pmatrix}=\begin{pmatrix}5\\ 4\end{pmatrix}\Rightarrow\begin{pmatrix}2 & 3\\ 1 & 1\end{pmatrix}^{-1}\begin{pmatrix}2 & 3\\ 1 & 1\end{pmatrix}\begin{pmatrix}x\\ y\end{pmatrix}=\begin{pmatrix}2 & 3\\ 1 & 1\end{pmatrix}^{-1}\begin{pmatrix}5\\ 4\end{pmatrix}$

$\Rightarrow\begin{pmatrix}1 & 0\\ 0 & 1\end{pmatrix}\begin{pmatrix}x\\ y\end{pmatrix}=\begin{pmatrix}2 & 3\\ 1 & 1\end{pmatrix}^{-1}\begin{pmatrix}5\\ 4\end{pmatrix}\Rightarrow\begin{pmatrix}x\\ y\end{pmatrix}=\begin{pmatrix}2 & 3\\ 1 & 1\end{pmatrix}^{-1}\begin{pmatrix}5\\ 4\end{pmatrix}$

$\begin{vmatrix}2 & 3\\ 1 & 1\end{vmatrix}=2\times1-3\times1=-1$

$\begin{pmatrix}2 & 3\\ 1 & 1\end{pmatrix}^{-1}=\frac{1}{-1}\begin{pmatrix}1 & -3\\ -1 & 2\end{pmatrix}=\begin{pmatrix}-1 & 3\\ 1 & -2\end{pmatrix}$

$\therefore\begin{pmatrix}x\\ y\end{pmatrix}=\begin{pmatrix}-1 & 3\\ 1 & -2\end{pmatrix}\begin{pmatrix}5\\ 4\end{pmatrix}=\begin{pmatrix}7\\ -3\end{pmatrix}$

So $x=7$ and $y=-3$

**44) b)**

$\begin{pmatrix}3 & 1\\ 1 & 2\end{pmatrix}\begin{pmatrix}x\\ y\end{pmatrix}=\begin{pmatrix}-3\\ 9\end{pmatrix}\Rightarrow\begin{pmatrix}3 & 1\\ 1 & 2\end{pmatrix}^{-1}\begin{pmatrix}3 & 1\\ 1 & 2\end{pmatrix}\begin{pmatrix}x\\ y\end{pmatrix}=\begin{pmatrix}3 & 1\\ 1 & 2\end{pmatrix}^{-1}\begin{pmatrix}-3\\ 9\end{pmatrix}$

$\Rightarrow\begin{pmatrix}1 & 0\\ 0 & 1\end{pmatrix}\begin{pmatrix}x\\ y\end{pmatrix}=\begin{pmatrix}3 & 1\\ 1 & 2\end{pmatrix}^{-1}\begin{pmatrix}-3\\ 9\end{pmatrix}\Rightarrow\begin{pmatrix}x\\ y\end{pmatrix}=\begin{pmatrix}3 & 1\\ 1 & 2\end{pmatrix}^{-1}\begin{pmatrix}-3\\ 9\end{pmatrix}$

$\begin{vmatrix}3 & 1\\ 1 & 2\end{vmatrix}=3\times2-1\times1=5$

$\begin{pmatrix}3 & 1\\ 1 & 2\end{pmatrix}^{-1}=\frac{1}{5}\begin{pmatrix}2 & -1\\ -1 & 3\end{pmatrix}=\begin{pmatrix}\frac{2}{5} & -\frac{1}{5}\\ -\frac{1}{5} & \frac{3}{5}\end{pmatrix}$

$\therefore\begin{pmatrix}x\\ y\end{pmatrix}=\begin{pmatrix}\frac{2}{5} & -\frac{1}{5}\\ -\frac{1}{5} & \frac{3}{5}\end{pmatrix}\begin{pmatrix}-3\\ 9\end{pmatrix}=\begin{pmatrix}-3\\ 6\end{pmatrix}$

So $x=-3$ and $y=6$

**44) c)**

$\begin{pmatrix}4 & -1\\ 8 & -3\end{pmatrix}\begin{pmatrix}x\\ y\end{pmatrix}=\begin{pmatrix}15\\ 1\end{pmatrix}\Rightarrow\begin{pmatrix}4 & -1\\ 8 & -3\end{pmatrix}^{-1}\begin{pmatrix}4 & -1\\ 8 & -3\end{pmatrix}\begin{pmatrix}x\\ y\end{pmatrix}=\begin{pmatrix}4 & -1\\ 8 & -3\end{pmatrix}^{-1}\begin{pmatrix}15\\ 1\end{pmatrix}$

$\Rightarrow\begin{pmatrix}1 & 0\\ 0 & 1\end{pmatrix}\begin{pmatrix}x\\ y\end{pmatrix}=\begin{pmatrix}4 & -1\\ 8 & -3\end{pmatrix}^{-1}\begin{pmatrix}15\\ 1\end{pmatrix}\Rightarrow\begin{pmatrix}x\\ y\end{pmatrix}=\begin{pmatrix}4 & -1\\ 8 & -3\end{pmatrix}^{-1}\begin{pmatrix}15\\ 1\end{pmatrix}$

$\begin{vmatrix}4 & -1\\ 8 & -3\end{vmatrix}=4\times(-3)-(-1)\times8=-4$

$\begin{pmatrix}4 & -1\\ 8 & -3\end{pmatrix}^{-1}=\frac{1}{-4}\begin{pmatrix}-3 & 1\\ -8 & 4\end{pmatrix}=\begin{pmatrix}\frac{3}{4} & -\frac{1}{4}\\ 2 & -1\end{pmatrix}$

$\therefore\begin{pmatrix}x\\ y\end{pmatrix}=\begin{pmatrix}\frac{3}{4} & -\frac{1}{4}\\ 2 & -1\end{pmatrix}\begin{pmatrix}15\\ 1\end{pmatrix}=\begin{pmatrix}11\\ 29\end{pmatrix}$

So $x=11$ and $y=29$

**44) d)**

$\begin{pmatrix}-6 & 2\\ 9 & -4\end{pmatrix}\begin{pmatrix}x\\ y\end{pmatrix}=\begin{pmatrix}-7\\ -10\end{pmatrix}\Rightarrow\begin{pmatrix}-6 & 2\\ 9 & -4\end{pmatrix}^{-1}\begin{pmatrix}2 & -6\\ 9 & -4\end{pmatrix}\begin{pmatrix}x\\ y\end{pmatrix}=\begin{pmatrix}-6 & 2\\ 9 & -4\end{pmatrix}^{-1}\begin{pmatrix}-7\\ -10\end{pmatrix}$

$\Rightarrow\begin{pmatrix}1 & 0\\ 0 & 1\end{pmatrix}\begin{pmatrix}x\\ y\end{pmatrix}=\begin{pmatrix}-6 & 2\\ 9 & -4\end{pmatrix}^{-1}\begin{pmatrix}-7\\ -10\end{pmatrix}\Rightarrow\begin{pmatrix}x\\ y\end{pmatrix}=\begin{pmatrix}-6 & 2\\ 9 & -4\end{pmatrix}^{-1}\begin{pmatrix}-7\\ -10\end{pmatrix}$

$\begin{vmatrix}-6 & 2\\ 9 & -4\end{vmatrix}=(-6)\times(-4)-2\times 9=6$

$\begin{pmatrix}-6 & 2\\ 9 & -4\end{pmatrix}^{-1}=\frac{1}{6}\begin{pmatrix}-4 & -2\\ -9 & -6\end{pmatrix}=\begin{pmatrix}-\frac{2}{3} & -\frac{1}{3}\\ -\frac{3}{2} & -1\end{pmatrix}$

$\therefore\begin{pmatrix}x\\ y\end{pmatrix}=\begin{pmatrix}-\frac{2}{3} & -\frac{1}{3}\\ -\frac{3}{2} & -1\end{pmatrix}\begin{pmatrix}-7\\ -10\end{pmatrix}=\begin{pmatrix}8\\ \frac{41}{2}\end{pmatrix}$

So $x=8$ and $y=\frac{41}{2}$

**45)**

$\begin{vmatrix}2 & 9k\\ k & 18\end{vmatrix}=2\times 18-9k\times k=36-9k^2$

No unique solution when $\begin{vmatrix}2 & 9k\\ k & 18\end{vmatrix}=0\Rightarrow 36-9k^2=0\Rightarrow k=\pm 2$

When $k=2$:

$2x+18y=6$

$2x+18y=6$

So the two lines are coincident when $k=2$

When $k=-2$:

$2x-18y=6$

$-2x+18y=6\Rightarrow 2x-18y=-6$

So the two lines are parallel when $k=-2$

**46)**

$\begin{vmatrix} 8k & 48 \\ 3 & 2k \end{vmatrix} = 8k \times 2k - 48 \times 3 = 16k^2 - 144$

No unique solution when $\begin{vmatrix} 8k & 48 \\ 3 & 2k \end{vmatrix} = 0 \Rightarrow 16k^2 - 144 = 0 \Rightarrow k = \pm 3$

When $k = 3$:

$24x + 48y = 5 \Rightarrow 3x + 6y = \frac{5}{8}$

$3x + 6y = 3$

So the two lines are parallel when $k = 3$

When $k = -3$:

$-24x + 48y = 5 \Rightarrow 3x - 6y = -\frac{5}{8}$

$3x - 6y = 3$

So the two lines are parallel when $k = -3$

**47) a)**

The coefficients of $x$, $y$ and $z$ match the elements of matrix $\mathbf{A}$ when we divide through by 10:

$0.2x + 0y - 0.3z = 4$

$0x - 0.2y + 0.3z = -6$

$0.1x - 0.2y - 0.1z = 9$

Given that we know that $\mathbf{A}^{-1}$ exists, the three simultaneous equations must have a unique solution (as $\det \mathbf{A} \neq 0$)

**47) b)**

$$\mathbf{A}\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 4 \\ -6 \\ 9 \end{pmatrix} \Rightarrow \mathbf{A}^{-1}\mathbf{A}\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \mathbf{A}^{-1}\begin{pmatrix} 4 \\ -6 \\ 9 \end{pmatrix} \Rightarrow \mathbf{I}\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \mathbf{A}^{-1}\begin{pmatrix} 4 \\ -6 \\ 9 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 8 & 6 & -6 \\ 3 & 1 & -6 \\ 2 & 4 & -4 \end{pmatrix}\begin{pmatrix} 4 \\ -6 \\ 9 \end{pmatrix} = \begin{pmatrix} -58 \\ -48 \\ -52 \end{pmatrix}$$

So $x = -58$, $y = -48$ and $z = -52$

**48) a)**

$$\begin{pmatrix}1&1&2\\2&1&1\\3&-1&4\end{pmatrix}\begin{pmatrix}x\\y\\z\end{pmatrix}=\begin{pmatrix}1\\3\\2\end{pmatrix}\Rightarrow\begin{pmatrix}1&1&2\\2&1&1\\3&-1&4\end{pmatrix}^{-1}\begin{pmatrix}1&1&2\\2&1&1\\3&-1&4\end{pmatrix}\begin{pmatrix}x\\y\\z\end{pmatrix}=\begin{pmatrix}1&1&2\\2&1&1\\3&-1&4\end{pmatrix}^{-1}\begin{pmatrix}1\\3\\2\end{pmatrix}$$

$$\Rightarrow\begin{pmatrix}1&0&0\\0&1&0\\0&0&1\end{pmatrix}\begin{pmatrix}x\\y\\z\end{pmatrix}=\begin{pmatrix}1&1&2\\2&1&1\\3&-1&4\end{pmatrix}^{-1}\begin{pmatrix}1\\3\\2\end{pmatrix}\Rightarrow\begin{pmatrix}x\\y\\z\end{pmatrix}=\begin{pmatrix}1&1&2\\2&1&1\\3&-1&4\end{pmatrix}^{-1}\begin{pmatrix}1\\3\\2\end{pmatrix}$$

*Calculator:*

$$\begin{pmatrix}1&1&2\\2&1&1\\3&-1&4\end{pmatrix}^{-1}=\begin{pmatrix}-\frac{1}{2}&\frac{3}{5}&\frac{1}{10}\\\frac{1}{2}&\frac{1}{5}&-\frac{3}{10}\\\frac{1}{2}&-\frac{2}{5}&\frac{1}{10}\end{pmatrix}$$

*By hand:*

Find the cofactor matrix:

$$\begin{pmatrix}5&-5&-5\\-6&-2&4\\-1&3&-1\end{pmatrix}$$

Find the transpose of the cofactor matrix:

$$\begin{pmatrix}5&-6&-1\\-5&-2&3\\-5&4&-1\end{pmatrix}$$

Find the determinant of the original matrix:

$$\begin{vmatrix}1&1&2\\2&1&1\\3&-1&4\end{vmatrix}=1\begin{vmatrix}1&1\\-1&4\end{vmatrix}-1\begin{vmatrix}2&1\\3&4\end{vmatrix}+2\begin{vmatrix}2&1\\3&-1\end{vmatrix}$$

$$=1\big(1\times4-1\times(-1)\big)-1(2\times4-1\times3)+2(2\times(-1)-1\times3)$$

$$=1\times5-1\times5+2\times-5=-10$$

Then:

$$\begin{pmatrix}1&1&2\\2&1&1\\3&-1&4\end{pmatrix}^{-1}=\frac{1}{-10}\begin{pmatrix}5&-6&-1\\-5&-2&3\\-5&4&-1\end{pmatrix}=\begin{pmatrix}-\frac{1}{2}&\frac{3}{5}&\frac{1}{10}\\\frac{1}{2}&\frac{1}{5}&-\frac{3}{10}\\\frac{1}{2}&-\frac{2}{5}&\frac{1}{10}\end{pmatrix}$$

$$\therefore\begin{pmatrix}x\\y\\z\end{pmatrix}=\begin{pmatrix}-\frac{1}{2}&\frac{3}{5}&\frac{1}{10}\\\frac{1}{2}&\frac{1}{5}&-\frac{3}{10}\\\frac{1}{2}&-\frac{2}{5}&\frac{1}{10}\end{pmatrix}\begin{pmatrix}1\\3\\2\end{pmatrix}=\begin{pmatrix}\frac{3}{2}\\\frac{1}{2}\\-\frac{1}{2}\end{pmatrix}$$

So $x=\frac{3}{2}$, $y=\frac{1}{2}$ and $z=-\frac{1}{2}$

**48) b)**

$$\begin{pmatrix}1 & 1 & 0\\ 1 & -1 & 1\\ 2 & 0 & 5\end{pmatrix}\begin{pmatrix}x\\ y\\ z\end{pmatrix}=\begin{pmatrix}5\\ 8\\ 3\end{pmatrix}$$

$$\Rightarrow\begin{pmatrix}1 & 1 & 0\\ 1 & -1 & 1\\ 2 & 0 & 5\end{pmatrix}^{-1}\begin{pmatrix}1 & 1 & 0\\ 1 & -1 & 1\\ 2 & 0 & 5\end{pmatrix}\begin{pmatrix}x\\ y\\ z\end{pmatrix}=\begin{pmatrix}1 & 1 & 0\\ 1 & -1 & 1\\ 2 & 0 & 5\end{pmatrix}^{-1}\begin{pmatrix}5\\ 8\\ 3\end{pmatrix}$$

$$\Rightarrow\begin{pmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{pmatrix}\begin{pmatrix}x\\ y\\ z\end{pmatrix}=\begin{pmatrix}1 & 1 & 0\\ 1 & -1 & 1\\ 2 & 0 & 5\end{pmatrix}^{-1}\begin{pmatrix}5\\ 8\\ 3\end{pmatrix}\Rightarrow\begin{pmatrix}x\\ y\\ z\end{pmatrix}=\begin{pmatrix}1 & 1 & 0\\ 1 & -1 & 1\\ 2 & 0 & 5\end{pmatrix}^{-1}\begin{pmatrix}5\\ 8\\ 3\end{pmatrix}$$

*Calculator:*

$$\begin{pmatrix}1 & 1 & 0\\ 1 & -1 & 1\\ 2 & 0 & 5\end{pmatrix}^{-1}=\begin{pmatrix}\frac{5}{8} & \frac{5}{8} & -\frac{1}{8}\\ \frac{3}{8} & -\frac{5}{8} & \frac{1}{8}\\ -\frac{1}{4} & -\frac{1}{4} & \frac{1}{4}\end{pmatrix}$$

*By hand:*

Find the cofactor matrix:

$$\begin{pmatrix}-5 & -3 & 2\\ -5 & 5 & 2\\ 1 & -1 & -2\end{pmatrix}$$

Find the transpose of the cofactor matrix:

$$\begin{pmatrix}-5 & -5 & 1\\ -3 & 5 & -1\\ 2 & 2 & -2\end{pmatrix}$$

Find the determinant of the original matrix:

$$\begin{vmatrix}1 & 1 & 0\\ 1 & -1 & 1\\ 2 & 0 & 5\end{vmatrix}=1\begin{vmatrix}-1 & 1\\ 0 & 5\end{vmatrix}-1\begin{vmatrix}1 & 1\\ 2 & 5\end{vmatrix}+0\begin{vmatrix}1 & -1\\ 2 & 0\end{vmatrix}$$

$$=1(-1\times5-1\times0)-1(1\times5-1\times2)+0(1\times0-(-1)\times2)$$

$$=1\times(-5)-1\times3+0=-8$$

Then:

$$\begin{pmatrix}1 & 1 & 0\\ 1 & -1 & 1\\ 2 & 0 & 5\end{pmatrix}^{-1}=\frac{1}{-8}\begin{pmatrix}-5 & -5 & 1\\ -3 & 5 & -1\\ 2 & 2 & -2\end{pmatrix}=\begin{pmatrix}\frac{5}{8} & \frac{5}{8} & -\frac{1}{8}\\ \frac{3}{8} & -\frac{5}{8} & \frac{1}{8}\\ -\frac{1}{4} & -\frac{1}{4} & \frac{1}{4}\end{pmatrix}$$

$$\therefore\begin{pmatrix}x\\ y\\ z\end{pmatrix}=\begin{pmatrix}\frac{5}{8} & \frac{5}{8} & -\frac{1}{8}\\ \frac{3}{8} & -\frac{5}{8} & \frac{1}{8}\\ -\frac{1}{4} & -\frac{1}{4} & \frac{1}{4}\end{pmatrix}\begin{pmatrix}5\\ 8\\ 3\end{pmatrix}=\begin{pmatrix}\frac{31}{4}\\ -\frac{11}{4}\\ -\frac{5}{2}\end{pmatrix}$$

So $x=\frac{31}{4}$, $y=-\frac{11}{4}$ and $z=-\frac{5}{2}$

**48) c)**

$$\begin{pmatrix} 0 & 1 & -3 \\ -1 & 0 & 4 \\ -1 & -1 & 1 \end{pmatrix}\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} -7 \\ 3 \\ -2 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 0 & 1 & -3 \\ -1 & 0 & 4 \\ -1 & -1 & 1 \end{pmatrix}^{-1}\begin{pmatrix} 0 & 1 & -3 \\ -1 & 0 & 4 \\ -1 & -1 & 1 \end{pmatrix}\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 & 1 & -3 \\ -1 & 0 & 4 \\ -1 & -1 & 1 \end{pmatrix}^{-1}\begin{pmatrix} -7 \\ 3 \\ -2 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 & 1 & -3 \\ -1 & 0 & 4 \\ -1 & -1 & 1 \end{pmatrix}^{-1}\begin{pmatrix} -7 \\ 3 \\ -2 \end{pmatrix} \Rightarrow \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 & 1 & -3 \\ -1 & 0 & 4 \\ -1 & -1 & 1 \end{pmatrix}^{-1}\begin{pmatrix} -7 \\ 3 \\ -2 \end{pmatrix}$$

*Calculator:*

$$\begin{pmatrix} 0 & 1 & -3 \\ -1 & 0 & 4 \\ -1 & -1 & 1 \end{pmatrix}^{-1} = \begin{pmatrix} -\frac{2}{3} & -\frac{1}{3} & -\frac{2}{3} \\ \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{6} & \frac{1}{6} & -\frac{1}{6} \end{pmatrix}$$

*By hand:*

Find the cofactor matrix:

$$\begin{pmatrix} 4 & -3 & 1 \\ 2 & -3 & -1 \\ 4 & 3 & 1 \end{pmatrix}$$

Find the transpose of the cofactor matrix:

$$\begin{pmatrix} 4 & 2 & 4 \\ -3 & -3 & 3 \\ 1 & -1 & 1 \end{pmatrix}$$

Find the determinant of the original matrix:

$$\begin{vmatrix} 0 & 1 & -3 \\ -1 & 0 & 4 \\ -1 & -1 & 1 \end{vmatrix} = 0\begin{vmatrix} 0 & 4 \\ -1 & 1 \end{vmatrix} - 1\begin{vmatrix} -1 & 4 \\ -1 & 1 \end{vmatrix} - 3\begin{vmatrix} -1 & 0 \\ -1 & -1 \end{vmatrix}$$

$$= 0\big(0 \times 1 - 4 \times (-1)\big) - 1\big((-1) \times 1 - 4 \times (-1)\big) - 3\big((-1) \times (-1) - 0 \times (-1)\big)$$

$$= 0 - 1 \times 3 - 3 \times 1 = -6$$

Then:

$$\begin{pmatrix} 0 & 1 & -3 \\ -1 & 0 & 4 \\ -1 & -1 & 1 \end{pmatrix}^{-1} = \frac{1}{-6}\begin{pmatrix} 4 & 2 & 4 \\ -3 & -3 & 3 \\ 1 & -1 & 1 \end{pmatrix} = \begin{pmatrix} -\frac{2}{3} & -\frac{1}{3} & -\frac{2}{3} \\ \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{6} & \frac{1}{6} & -\frac{1}{6} \end{pmatrix}$$

$$\therefore \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} -\frac{2}{3} & -\frac{1}{3} & -\frac{2}{3} \\ \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{6} & \frac{1}{6} & -\frac{1}{6} \end{pmatrix}\begin{pmatrix} -7 \\ 3 \\ -2 \end{pmatrix} = \begin{pmatrix} 5 \\ -1 \\ 2 \end{pmatrix}$$

So $x = 5$ , $y = -1$ and $z = 2$

**49)**

$$\begin{pmatrix} 2 & 1 & 2 & -3 \\ -1 & 4 & 3 & 5 \\ 6 & -2 & 1 & -8 \\ -9 & -7 & 3 & 2 \end{pmatrix}\begin{pmatrix} a \\ b \\ c \\ d \end{pmatrix} = \begin{pmatrix} 8 \\ 14 \\ 11 \\ 6 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 2 & 1 & 2 & -3 \\ -1 & 4 & 3 & 5 \\ 6 & -2 & 1 & -8 \\ -9 & -7 & 3 & 2 \end{pmatrix}^{-1}\begin{pmatrix} 2 & 1 & 2 & -3 \\ -1 & 4 & 3 & 5 \\ 6 & -2 & 1 & -8 \\ -9 & -7 & 3 & 2 \end{pmatrix}\begin{pmatrix} a \\ b \\ c \\ d \end{pmatrix} = \begin{pmatrix} 2 & 1 & 2 & -3 \\ -1 & 4 & 3 & 5 \\ 6 & -2 & 1 & -8 \\ -9 & -7 & 3 & 2 \end{pmatrix}^{-1}\begin{pmatrix} 8 \\ 14 \\ 11 \\ 6 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}\begin{pmatrix} a \\ b \\ c \\ d \end{pmatrix} = \begin{pmatrix} 2 & 1 & 2 & -3 \\ -1 & 4 & 3 & 5 \\ 6 & -2 & 1 & -8 \\ -9 & -7 & 3 & 2 \end{pmatrix}^{-1}\begin{pmatrix} 8 \\ 14 \\ 11 \\ 6 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} a \\ b \\ c \\ d \end{pmatrix} = \begin{pmatrix} 2 & 1 & 2 & -3 \\ -1 & 4 & 3 & 5 \\ 6 & -2 & 1 & -8 \\ -9 & -7 & 3 & 2 \end{pmatrix}^{-1}\begin{pmatrix} 8 \\ 14 \\ 11 \\ 6 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} a \\ b \\ c \\ d \end{pmatrix} = \begin{pmatrix} -\frac{1}{2} & \frac{143}{578} & \frac{97}{289} & -\frac{15}{578} \\ \frac{1}{2} & -\frac{91}{578} & -\frac{88}{289} & -\frac{43}{578} \\ 0 & \frac{60}{289} & \frac{43}{289} & \frac{22}{289} \\ -\frac{1}{2} & \frac{145}{578} & \frac{64}{289} & \frac{5}{578} \end{pmatrix}\begin{pmatrix} 8 \\ 14 \\ 11 \\ 6 \end{pmatrix} = \begin{pmatrix} 3 \\ -2 \\ 5 \\ 2 \end{pmatrix}$$

So $a = 3$, $b = -2$, $c = 5$ and $d = 2$

**50) a)**

Find the cofactor matrix:

$$\begin{pmatrix} 0 & 1-k & 0 \\ -k & k^2-1 & 1 \\ 1 & 1-k & -1 \end{pmatrix}$$

Find the transpose of the cofactor matrix:

$$\begin{pmatrix} 0 & -k & 1 \\ 1-k & k^2-1 & 1-k \\ 0 & 1 & -1 \end{pmatrix}$$

Find the determinant of the original matrix:

$$\begin{vmatrix} k & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 0 & k \end{vmatrix} = k\begin{vmatrix} 0 & 1 \\ 0 & k \end{vmatrix} - 1\begin{vmatrix} 1 & 1 \\ 1 & k \end{vmatrix} + 1\begin{vmatrix} 1 & 0 \\ 1 & 0 \end{vmatrix}$$

$$= k(0 \times k - 1 \times 0) - 1(1 \times k - 1 \times 1) + 1(1 \times 0 - 0 \times 1)$$

$$= k \times 0 - 1 \times (k-1) + 1 \times 0 = 1 - k$$

Then:

$$\begin{pmatrix} k & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 0 & k \end{pmatrix}^{-1} = \frac{1}{1-k}\begin{pmatrix} 0 & -k & 1 \\ 1-k & k^2-1 & 1-k \\ 0 & 1 & -1 \end{pmatrix} = \begin{pmatrix} 0 & -\frac{k}{1-k} & \frac{1}{1-k} \\ 1 & \frac{(k-1)(k+1)}{1-k} & 1 \\ 0 & \frac{1}{1-k} & -\frac{1}{1-k} \end{pmatrix}$$

$$= \begin{pmatrix} 0 & -\frac{k}{1-k} & \frac{1}{1-k} \\ 1 & -k-1 & 1 \\ 0 & \frac{1}{1-k} & -\frac{1}{1-k} \end{pmatrix}$$

**50) b)**

When $k = 3$, $\mathbf{A}^{-1} = \begin{pmatrix} 0 & \frac{3}{2} & -\frac{1}{2} \\ 1 & -4 & 1 \\ 0 & -\frac{1}{2} & \frac{1}{2} \end{pmatrix}$

$$\mathbf{A}\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} -6 \\ 18 \\ -12 \end{pmatrix} \Rightarrow \mathbf{A}^{-1}\mathbf{A}\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \mathbf{A}^{-1}\begin{pmatrix} -6 \\ 18 \\ -12 \end{pmatrix} \Rightarrow \mathbf{I}\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \mathbf{A}^{-1}\begin{pmatrix} -6 \\ 18 \\ -12 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 & \frac{3}{2} & -\frac{1}{2} \\ 1 & -4 & 1 \\ 0 & -\frac{1}{2} & \frac{1}{2} \end{pmatrix}\begin{pmatrix} -6 \\ 18 \\ -12 \end{pmatrix} = \begin{pmatrix} 33 \\ -90 \\ -15 \end{pmatrix}$$

So $x = 33$ , $y = -90$ and $z = -15$

**51) a)**

Find the cofactor matrix:

$$\begin{pmatrix} 5 & 2k-3 & 2-3k \\ 0 & -3k-15 & 2k+10 \\ -5 & 13 & -17 \end{pmatrix}$$

Find the transpose of the cofactor matrix:

$$\begin{pmatrix} 5 & 0 & -5 \\ 2k-3 & -3k-15 & 13 \\ 2-3k & 2k+10 & -17 \end{pmatrix}$$

Find the determinant of the original matrix:

$$\begin{vmatrix} -5 & 2 & 3 \\ 1 & 3 & 2 \\ k & 2 & 3 \end{vmatrix} = -5\begin{vmatrix} 3 & 2 \\ 2 & 3 \end{vmatrix} - 2\begin{vmatrix} 1 & 2 \\ k & 3 \end{vmatrix} + 3\begin{vmatrix} 1 & 3 \\ k & 2 \end{vmatrix}$$

$$= -5(3\times 3 - 2\times 2) - 2(1\times 3 - 2\times k) + 3(1\times 2 - 3\times k)$$

$$= -5\times 5 - 2\times(3-2k) + 3\times(2-3k) = -5k - 25$$

Then:

$$\begin{pmatrix} -5 & 2 & 3 \\ 1 & 3 & 2 \\ k & 2 & 3 \end{pmatrix}^{-1} = \frac{1}{-5k-25}\begin{pmatrix} 5 & 0 & -5 \\ 2k-3 & -3k-15 & 13 \\ 2-3k & 2k+10 & -17 \end{pmatrix}$$

$$= \begin{pmatrix} \frac{5}{-5k-25} & 0 & -\frac{5}{-5k-25} \\ \frac{2k-3}{-5k-25} & \frac{-3k-15}{-5k-25} & \frac{13}{-5k-25} \\ \frac{2-3k}{-5k-25} & \frac{2k+10}{-5k-25} & -\frac{17}{-5k-25} \end{pmatrix} = \begin{pmatrix} -\frac{1}{k+5} & 0 & \frac{1}{k+5} \\ -\frac{2k-3}{5k+25} & \frac{3}{5} & -\frac{13}{5k+25} \\ -\frac{2-3k}{5k+25} & -\frac{2}{5} & \frac{17}{5k+25} \end{pmatrix}$$

**51) b)**

When $k = 5$, $\mathbf{A}^{-1} = \begin{pmatrix} -\frac{1}{10} & 0 & \frac{1}{10} \\ -\frac{7}{50} & \frac{3}{5} & -\frac{13}{50} \\ \frac{13}{50} & -\frac{2}{5} & \frac{17}{50} \end{pmatrix}$

$$\mathbf{A}\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 2 \\ -5 \\ 9 \end{pmatrix} \Rightarrow \mathbf{A}^{-1}\mathbf{A}\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \mathbf{A}^{-1}\begin{pmatrix} 2 \\ -5 \\ 9 \end{pmatrix} \Rightarrow \mathbf{I}\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \mathbf{A}^{-1}\begin{pmatrix} 2 \\ -5 \\ 9 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} -\frac{1}{10} & 0 & \frac{1}{10} \\ \frac{7}{50} & \frac{3}{5} & -\frac{13}{50} \\ \frac{13}{50} & -\frac{2}{5} & \frac{17}{50} \end{pmatrix}\begin{pmatrix} 2 \\ -5 \\ 9 \end{pmatrix} = \begin{pmatrix} \frac{7}{10} \\ -\frac{281}{50} \\ \frac{279}{50} \end{pmatrix}$$

So $x = \frac{7}{10}$, $y = -\frac{281}{50}$ and $z = \frac{279}{50}$

**52) a)**

$$\mathbf{C}^{-1}\mathbf{C}\mathbf{A}\mathbf{B}^{-1}\mathbf{B} = (\mathbf{C}^{-1}\mathbf{C})\mathbf{A}(\mathbf{B}^{-1}\mathbf{B}) = \mathbf{I}\mathbf{A}\mathbf{I} = \mathbf{A}$$

**52) b)**
$\mathbf{B(AB)^{-1} = B(B^{-1}A^{-1}) = (BB^{-1})A^{-1} = IA^{-1} = A^{-1}}$

**52) c)**
$\mathbf{(CA^{-1})^{-1}C = (AC^{-1})C = A(C^{-1}C) = AI = A}$

**52) d)**
$\mathbf{C(ABC)^{-1}A = C(C^{-1}B^{-1}A^{-1})A = (CC^{-1})B^{-1}(A^{-1}A) = IB^{-1}I = B^{-1}}$

**52) e)**
$\mathbf{(B^{-1}A)^{-1}(C^{-1}B)^{-1} = (A^{-1}B)(B^{-1}C) = A^{-1}(BB^{-1})C = A^{-1}IC = A^{-1}C}$

**52) f)**
$\mathbf{(AB^{-1}C)^{-1}(C^{-1}BA^{-1})^{-1} = (C^{-1}BA^{-1})(AB^{-1}C) = C^{-1}B(A^{-1}A)B^{-1}C}$
$\mathbf{= C^{-1}BIB^{-1}C = C^{-1}(BB^{-1})C = C^{-1}IC = C^{-1}C = I}$

**52) g)**
$\mathbf{(ACBA^{-1})^{-1}(AB^{-1}C^{-1}A^{-1})^{-1} = (AB^{-1}C^{-1}A^{-1})(ACBA^{-1})}$
$\mathbf{= AB^{-1}C^{-1}(A^{-1}A)CBA^{-1} = AB^{-1}C^{-1}ICBA^{-1} = AB^{-1}(C^{-1}C)BA^{-1}}$
$\mathbf{= AB^{-1}IBA^{-1} = A(B^{-1}B)A^{-1} = AIA^{-1} = AA^{-1} = I}$

**53)**
$\mathbf{AB = C \Rightarrow A^{-1}AB = A^{-1}C \Rightarrow IB = A^{-1}C \Rightarrow B = A^{-1}C}$

**54)**
$\mathbf{ABC = A \Rightarrow A^{-1}ABC = A^{-1}A \Rightarrow IBC = I \Rightarrow BC = I}$
$\mathbf{\Rightarrow BCC^{-1} = IC^{-1} \Rightarrow BI = C^{-1} \Rightarrow B = C^{-1}}$

**55) a)**
Matrix multiplication is not generally commutative, so equating $\mathbf{B^{-1}ABA^{-1}}$ to $\mathbf{B^{-1}BAA^{-1}}$ is incorrect (i.e. $\mathbf{AB \neq BA}$)

**55) b)**
$\mathbf{(B^{-1}A)^{-1} = A^{-1}B}$

**55) c)**
$\mathbf{(B^{-1}A)(B^{-1}A)^{-1} = I \Rightarrow BB^{-1}A(B^{-1}A)^{-1} = BI \Rightarrow IA(B^{-1}A)^{-1} = B \Rightarrow A(B^{-1}A)^{-1} = B}$
$\mathbf{\Rightarrow A^{-1}A(B^{-1}A)^{-1} = A^{-1}B \Rightarrow I(B^{-1}A)^{-1} = A^{-1}B \Rightarrow (B^{-1}A)^{-1} = A^{-1}B}$

**56)**

$\mathbf{B} = \mathbf{A}^{-1}\mathbf{CA} \Rightarrow \mathbf{AB} = \mathbf{AA}^{-1}\mathbf{CA} \Rightarrow \mathbf{AB} = \mathbf{ICA} \Rightarrow \mathbf{AB} = \mathbf{CA} \Rightarrow \mathbf{ABA}^{-1} = \mathbf{CAA}^{-1}$
$\Rightarrow \mathbf{ABA}^{-1} = \mathbf{CI} \Rightarrow \mathbf{C} = \mathbf{ABA}^{-1}$

**57)**

$\mathbf{ABA} = \mathbf{I} \Rightarrow \mathbf{A}^{-1}\mathbf{ABA} = \mathbf{A}^{-1}\mathbf{I} \Rightarrow \mathbf{IBA} = \mathbf{A}^{-1} \Rightarrow \mathbf{BA} = \mathbf{A}^{-1}$
$\Rightarrow \mathbf{BAA}^{-1} = \mathbf{A}^{-1}\mathbf{A}^{-1} \Rightarrow \mathbf{BI} = \mathbf{A}^{-1}\mathbf{A}^{-1} \Rightarrow \mathbf{B} = \mathbf{A}^{-1}\mathbf{A}^{-1}$

**58)**

$(\mathbf{AB})(\mathbf{AB})^{-1} = \mathbf{I} \Rightarrow \mathbf{A}^{-1}\mathbf{AB}(\mathbf{AB})^{-1} = \mathbf{A}^{-1}\mathbf{I} \Rightarrow \mathbf{IB}(\mathbf{AB})^{-1} = \mathbf{A}^{-1} \Rightarrow \mathbf{B}(\mathbf{AB})^{-1} = \mathbf{A}^{-1}$
$\Rightarrow \mathbf{B}^{-1}\mathbf{B}(\mathbf{AB})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1} \Rightarrow \mathbf{I}(\mathbf{AB})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1} \Rightarrow (\mathbf{AB})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1}$

**59)**

$(\mathbf{ABC})(\mathbf{ABC})^{-1} = \mathbf{I} \Rightarrow \mathbf{A}^{-1}\mathbf{ABC}(\mathbf{ABC})^{-1} = \mathbf{A}^{-1}\mathbf{I} \Rightarrow \mathbf{IBC}(\mathbf{ABC})^{-1} = \mathbf{A}^{-1}$
$\Rightarrow \mathbf{BC}(\mathbf{ABC})^{-1} = \mathbf{A}^{-1} \Rightarrow \mathbf{B}^{-1}\mathbf{BC}(\mathbf{ABC})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1} \Rightarrow \mathbf{IC}(\mathbf{ABC})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1}$
$\Rightarrow \mathbf{C}(\mathbf{ABC})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1} \Rightarrow \mathbf{C}^{-1}\mathbf{C}(\mathbf{ABC})^{-1} = \mathbf{C}^{-1}\mathbf{B}^{-1}\mathbf{A}^{-1}$
$\Rightarrow \mathbf{I}(\mathbf{ABC})^{-1} = \mathbf{C}^{-1}\mathbf{B}^{-1}\mathbf{A}^{-1} \Rightarrow (\mathbf{ABC})^{-1} = \mathbf{C}^{-1}\mathbf{B}^{-1}\mathbf{A}^{-1}$

**60)**

$(\mathbf{A}^{-1}\mathbf{B}^{-1}\mathbf{C}^{-1})(\mathbf{A}^{-1}\mathbf{B}^{-1}\mathbf{C}^{-1})^{-1} = \mathbf{I} \Rightarrow \mathbf{AA}^{-1}\mathbf{B}^{-1}\mathbf{C}^{-1}(\mathbf{A}^{-1}\mathbf{B}^{-1}\mathbf{C}^{-1})^{-1} = \mathbf{AI}$
$\Rightarrow \mathbf{IB}^{-1}\mathbf{C}^{-1}(\mathbf{A}^{-1}\mathbf{B}^{-1}\mathbf{C}^{-1})^{-1} = \mathbf{A} \Rightarrow \mathbf{B}^{-1}\mathbf{C}^{-1}(\mathbf{A}^{-1}\mathbf{B}^{-1}\mathbf{C}^{-1})^{-1} = \mathbf{A}$
$\Rightarrow \mathbf{BB}^{-1}\mathbf{C}^{-1}(\mathbf{A}^{-1}\mathbf{B}^{-1}\mathbf{C}^{-1})^{-1} = \mathbf{BA} \Rightarrow \mathbf{IC}^{-1}(\mathbf{A}^{-1}\mathbf{B}^{-1}\mathbf{C}^{-1})^{-1} = \mathbf{BA}$
$\Rightarrow \mathbf{C}^{-1}(\mathbf{A}^{-1}\mathbf{B}^{-1}\mathbf{C}^{-1})^{-1} = \mathbf{BA} \Rightarrow \mathbf{CC}^{-1}(\mathbf{A}^{-1}\mathbf{B}^{-1}\mathbf{C}^{-1})^{-1} = \mathbf{CBA}$
$\Rightarrow \mathbf{I}(\mathbf{A}^{-1}\mathbf{B}^{-1}\mathbf{C}^{-1})^{-1} = \mathbf{CBA} \Rightarrow (\mathbf{A}^{-1}\mathbf{B}^{-1}\mathbf{C}^{-1})^{-1} = \mathbf{CBA}$

**61)**

$\det(\mathbf{AB}) = \det\mathbf{A} \times \det\mathbf{B}$
If $\mathbf{B} = \mathbf{A}^{-1}$, then
$\det(\mathbf{AA}^{-1}) = \det\mathbf{A} \times \det(\mathbf{A}^{-1}) \Rightarrow \det(\mathbf{I}) = \det\mathbf{A} \times \det(\mathbf{A}^{-1})$
$\Rightarrow 1 = \det\mathbf{A} \times \det(\mathbf{A}^{-1}) \Rightarrow \det(\mathbf{A}^{-1}) = \frac{1}{\det\mathbf{A}}$

**62)**

$(\mathbf{AA}^{-1})^{\mathrm{T}} = (\mathbf{A}^{-1})^{\mathrm{T}}\mathbf{A}^{\mathrm{T}} \Rightarrow \mathbf{I}^{\mathrm{T}} = (\mathbf{A}^{-1})^{\mathrm{T}}\mathbf{A}^{\mathrm{T}} \Rightarrow \mathbf{I} = (\mathbf{A}^{-1})^{\mathrm{T}}\mathbf{A}^{\mathrm{T}}$
$\Rightarrow \mathbf{I}(\mathbf{A}^{\mathrm{T}})^{-1} = (\mathbf{A}^{-1})^{\mathrm{T}}\mathbf{A}^{\mathrm{T}}(\mathbf{A}^{\mathrm{T}})^{-1}$
$\Rightarrow (\mathbf{A}^{\mathrm{T}})^{-1} = (\mathbf{A}^{-1})^{\mathrm{T}}\mathbf{I} \Rightarrow (\mathbf{A}^{\mathrm{T}})^{-1} = (\mathbf{A}^{-1})^{\mathrm{T}}$

**63) a)**

$$\mathbf{AB} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \times \frac{1}{ad-bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix} = \frac{1}{ad-bc}\begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

$$= \frac{1}{ad-bc}\begin{pmatrix} ad-bc & -ab+ab \\ cd-cd & ad-bc \end{pmatrix} = \frac{1}{ad-bc}\begin{pmatrix} ad-bc & 0 \\ 0 & ad-bc \end{pmatrix}$$

$$= \begin{pmatrix} \frac{ad-bc}{ad-bc} & 0 \\ 0 & \frac{ad-bc}{ad-bc} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \mathbf{I}$$

**63) b)**

$$\mathbf{BA} = \frac{1}{ad-bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}\begin{pmatrix} a & b \\ c & d \end{pmatrix} = \frac{1}{ad-bc}\begin{pmatrix} ad-bc & bd-bd \\ -ac+ac & ad-bc \end{pmatrix}$$

$$= \frac{1}{ad-bc}\begin{pmatrix} ad-bc & 0 \\ 0 & ad-bc \end{pmatrix} = \begin{pmatrix} \frac{ad-bc}{ad-bc} & 0 \\ 0 & \frac{ad-bc}{ad-bc} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \mathbf{I}$$

**64)**

Let $\mathbf{A}^{-1} = \begin{pmatrix} p & q \\ r & s \end{pmatrix}$

$$\mathbf{AA}^{-1} = \mathbf{I} \Rightarrow \begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} p & q \\ r & s \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \Rightarrow \begin{pmatrix} ap+br & aq+bs \\ cp+dr & cq+ds \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$ap + br = 1$ (1)

$aq + bs = 0$ (2)

$cp + dr = 0$ (3)

$cq + ds = 1$ (4)

(1): $p = \frac{1-br}{a}$

(3): $p = -\frac{dr}{c}$

$\therefore \frac{1-br}{a} = -\frac{dr}{c} \Rightarrow c - bcr = -adr \Rightarrow adr - bcr = -c \Rightarrow (ad-bc)r = -c \Rightarrow r = -\frac{c}{ad-bc}$

$\therefore p = -\frac{d}{c} \times -\frac{c}{ad-bc} = \frac{d}{ad-bc}$

(2): $q = -\frac{bs}{a}$

(4): $q = \frac{1-ds}{c}$

$\therefore \frac{1-ds}{c} = -\frac{bs}{a} \Rightarrow a - ads = -bcs \Rightarrow a = ads - bcs \Rightarrow a = (ad-bc)s \Rightarrow s = \frac{a}{ad-bc}$

$\therefore q = -\frac{b}{a} \times \frac{a}{ad-bc} = -\frac{b}{ad-bc}$

So:

$$\mathbf{A}^{-1} = \begin{pmatrix} p & q \\ r & s \end{pmatrix} = \begin{pmatrix} \frac{d}{ad-bc} & -\frac{b}{ad-bc} \\ -\frac{c}{ad-bc} & \frac{a}{ad-bc} \end{pmatrix} = \frac{1}{ad-bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$